05.

Lowest Common Multiple & Highest Common Factor

Type - 1

1. The LCM of 14, 42 and 77 is:

(a) 168 (b) 308
(c) 462 (d) 154

RRB RPF Constable -18/01/2019 (Shift-I)

Ans : (c) Finding the LCM by common division method,

2	14,	42,	77
3	7,	21,	77
7	7,	7,	11
11	1,	1,	11
	1,	1,	1

L.C.M. = 2 ×3 × 7 × 11
= 6 × 77
= 462

2. Find the LCM of ab^2c^2, a^2bc and $a^3b^3c^2$

(a) $a^2b^2c^2$ (b) abc
(c) $a^3b^3c^2$ (d) $a^3b^3c^3$

RRB JE - 23/05/2019 (Shift-II)

Ans : (c) Finding the LCM by using prime factorization method,

$ab^2c^2 = a\times b\times b\times c\times c$
$a^2bc = a\times a\times b\times c$
$a^3b^3c^2 = a\times a\times a\times b\times b\times b\times c\times c$
LCM $= a^3b^3c^2$

3. Find the LCM of $(2^2 \times 3^2 \times 5 \times 7)$, $(2^2\times3\times5^2\times7)$ and $(2 \times 3 \times 5 \times 7)$.

(a) 6300 (b) 7200
(c) 9000 (d) 8400

RRB JE - 24/05/2019 (Shift-III)

Ans : (a) Finding the LCM by using prime factorization method,

$2^2 \times 3^2 \times 5 \times 7 = 1260$
$2^2 \times 3 \times 5^2 \times 7 = 2100$
$2 \times 3 \times 5 \times 7 = 210$
LCM = $2 \times 2 \times 3 \times 3 \times 5 \times 5 \times 7 = 6300$

4. Find the LCM of 60, 120 and 225.

(a) 360 (b) 1800
(c) 600 (d) 900

RRB RPF SI -06/01/2019 (Shift-II)

Ans : (b) Finding the LCM by using common division method,

2	60,	120,	225
2	30,	60,	225
2	15,	30,	225
3	15,	15,	225
3	5,	5,	75
5	5,	5,	25
5	1,	1,	5
	1,	1,	1

The required LCM
$= 2 \times 2 \times 2 \times 3 \times3\times 5 \times 5 = 1800$

5. Find the LCM of 6 (xy – y), $8(x^4y–xy)$.

(a) 24 xy $(x^3–1)$ (b) $24(x^3–1)$
(c) 24 xy (d) $(x^3– 1)$ xy

RRB JE - 26/05/2019 (Shift-III)

Ans : (a) Finding the LCM by prime factorization method,

$6(xy–y) = 6y (x–1) = 2\times3\times y (x–1)$
$8(x^4y – xy) = 8xy (x^3 –1)$
$= 8xy (x–1) (x^2+x+1)$
$\therefore$ LCM $= 2\times2\times2\times3xy (x–1) (x^2 + x+1)$
$= 24xy (x^3–1)$

6. The LCM of 2.05, 1.05, 2 is equals to what among the following?

(a) LCM of 205/100, 105/100 and 200/100
(b) LCM of 205, 105 and 200
(c) LCM of 21/20, 41/20 and 20/20
(d) LCM of 205, 105 and 200/10

RRB JE - 30/05/2019 (Shift-II)

Ans : (a) From the question,

$$2.05 = \frac{205}{100}$$

$$1.05 = \frac{105}{100}$$

$$2 = \frac{200}{100}$$

So, the LCM of 2.05, 1.05, 2

$$= \text{LCM of } \left(\frac{205}{100}, \frac{105}{100}, \frac{200}{100}\right)$$

7. The LCM of 36 and K is 72. Find the possible value of K.

(a) 24 only (b) 8, 24, 72
(c) 24, 72 (d) 8 only

RRB JE - 27/06/2019 (Shift-III)

Ans : (b) Given,
The LCM of 36 and K = 72.
From options,
The possible value of K will be = 8, 24 and 72.

8. The LCM of 112, 72 and 90 is?

(a) 3780 (b) 7560
(c) 2520 (d) 5040

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (d) : By using prime factorization method,
$112 = 2 \times 2 \times 2 \times 2 \times 7$
$72 = 2 \times 2 \times 2 \times 3 \times 3$
$90 = 2 \times 3 \times 3 \times 5$
$\therefore$ LCM $= 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7 = 5040$

9. What is the LCM of 72 and 84?

(a) $12 \times 6 \times 7$ (b) $6 \times 2 \times 7$
(c) $12 \times 7 \times 3$ (d) 72×84

RRB Group-D – 03/10/2018 (Shift-I)

Ans : (a) By using prime factorization method,
$72 \Rightarrow 2 \times 2 \times 2 \times 3 \times 3$
$84 \Rightarrow 2 \times 2 \times 3 \times 7$
LCM $= 2 \times 2 \times 2 \times 3 \times 3 \times 7 = 12 \times 6 \times 7$

10. What is the LCM of the two corresponding numbers, x and y?

(a) X Y (b) $\frac{(X.Y)}{2}$
(c) 1 (d) $\frac{X}{Y}$

RRB Group-D – 05/10/2018 (Shift-II)

Ans : (a) The LCM of the numbers x and y = x y
Note:- The LCM of the corresponding numbers is its product.

11. The LCM of 36, 27 and 72 is:

(a) 108 (b) 432
(c) 324 (d) 216

RRB Group-D – 31/10/2018 (Shift-II)

Ans : (d) Finding the LCM by using common division method,

2	36,	27,	72
2	18,	27,	36
2	9,	27,	18
3	9,	27,	9
3	3,	9,	3
3	1,	3,	1
	1,	1,	1

Hence, the required LCM = $2\times2\times2\times3\times3\times3 = 216$

12. The LCM of 16, 24, 36, 52 and 54 is:

(a) 5616 (b) 5216
(c) 432 (d) 5618

RRB Paramedical Exam – 20/07/2018 (Shift-I)

Ans : (a) Finding the LCM by using common division method,

2	16,	24,	36,	52,	54
2	8,	12,	18,	26,	27
2	4,	6,	9,	13,	27
2	2,	3,	9,	13,	27
3	1,	3,	9,	13,	27
3	1,	1,	3,	13,	9
3	1,	1,	1,	13,	3
13	1,	1,	1,	13,	1
	1,	1,	1,	1,	1

Hence, the required LCM
$= 2\times2\times2\times2\times3\times3\times3\times13 = 5616$

13. Find the LCM of 20, 40, 25 and 30:

(a) 800 (b) 500
(c) 600 (d) 400

RRB Group-D – 10/10/2018 (Shift-III)

Ans : (c) Finding the LCM by using common division method,

2	20,	40,	25,	30
2	10,	20,	25,	15
2	5,	10,	25,	15
3	5,	5,	25,	15
5	5,	5,	25,	5
5	1,	1,	5,	1
	1,	1,	1,	1

Hence, the required LCM = $2\times2\times2\times3\times5\times5 = 600$

14. The HCF of 290 and 660 is 10. What is their LCM?

(a) 38280 (b) 9570
(c) 19140 (d) 191400

RRB Group-D – 10/11/2018 (Shift-I)

Ans : (c) As we know,
First number $\times$ second number = LCM $\times$ HCF
$290 \times 660 = \text{LCM} \times 10$
$\text{LCM} = 290 \times 66$
LCM = 19140

15. What is the LCM of 21 and 280?

(a) 560 (b) 1120
(c) 840 (d) 1680

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (c) Finding the LCM by using common division method,

2	21, 280
2	21, 140
2	21, 70
3	21, 35
5	7, 35
7	7, 7
	1, 1

Hence, the required LCM = $2\times2\times2\times3\times5\times7 = 840$

16. What is the LCM of 120 and 280?

(a) 840 (b) 1680
(c) 1120 (d) 720

RRB Group-D – 12/12/2018 (Shift-I)

Ans. (a) Finding the LCM by using common division method,

2	120, 280
2	60, 140
2	30, 70
3	15, 35
5	5, 35
7	1, 7
	1, 1

Hence, the required LCM = 2×2×2×3×5×7 = 840

17. What is the LCM of 94, 188 and 235?

(a) 940 (b) 705
(c) 470 (d) 1880

RRB Group-D – 15/10/2018 (Shift-II)

Ans : (a) Finding the LCM by using common division method,

2	94, 188, 235
2	47, 94, 235
5	47, 47, 235
47	47, 47, 47
	1, 1, 1

LCM = 2 × 2 × 5 × 47 = 940

18. The LCM of $15x^3y^4$ and $12x^2y^5$ is:

(a) $60x^3y^5$ (b) $25x^3y^5$
(c) $12x^3y^3$ (d) $15x^3y^5$

RRB Group-D – 23/09/2018 (Shift-II)

Ans : (a) Finding the LCM by using prime factorization method,

$$15x^3y^4 = 3\times5\times x^3\times y^4$$

$$12x^2y^5 = 2\times2\times3\times x^2\times y^5$$

So, the required LCM

$$= 2\times2\times3\times5\times x^3\times y^5 = 60x^3y^5$$

19. Find the LCM of 24, 96 and 36.

(a) 576 (b) 216
(c) 288 (d) 144

RRB NTPC 17.01.2017 Shift-2

Ans : (c) Finding the LCM by using common division method,

2	24,	96,	36
2	12,	48,	18
2	6,	24,	9
2	3,	12,	9
2	3,	6,	9
3	3,	3,	9
3	1,	1,	3
	1,	1,	1

The required LCM = 2 × 2 × 2 × 2 × 2 × 3 × 3 = 288

20. Find the LCM of 15, 25 and 29.

(a) 2335 (b) 3337
(c) 2175 (d) 2375

RRB NTPC 02.04.2016 Shift : 2

Ans : (c) Finding the LCM by using prime factorization method,

15 → 1 × 3 × 5
25 → 1 × 5 × 5
29 → 1 × 29

Hence, the required LCM = 1 × 3 × 5 × 5 × 29 = 2175

21. Find the LCM of 18, 33 and 37?

(a) 2442 (b) 7326
(c) 814 (d) 1221

RRB NTPC 28.04.2016 Shift : 2

Ans : (b) Finding the LCM by using common division method,

3	18,	33,	37
	6,	11,	37

∴ Hence, the LCM = 3 × 6 × 11 × 37 = 7326

22. The LCM of 14, 35 and 56 is:

(a) 280 (b) 140
(c) 210 (d) 560

RRB ALP & Tec. (29-08-18 Shift-I)

Ans : (a) Finding the LCM by using prime factorization method,

14 = 2 × 7
35 = 5 × 7
56 = 2 × 2 × 2 × 7

Hence, the required LCM = 7 × 5 × 2 × 2 × 2 = 280

23. Find the LCM of 34, 51 and 68.

(a) 238 (b) 204
(c) 136 (d) 102

RRB ALP & Tec. (21-08-18 Shift-II)

Ans : (b) Finding the LCM by using prime factorization method,

34 = 2 × 17
51 = 3 × 17
68 = 2 × 2 × 17

Hence, the required LCM = 2 × 2 × 3 × 17 = 204

24. The LCM of 15, 18 and 24 is:

(a) 540 (b) 3
(c) 6 (d) 360

RRB ALP & Tec. (20-08-18 Shift-I)

Ans : (d) Finding the LCM by using prime factorization method,

15 = 3 × 5, 18 = 2 × 3 × 3,
24 = 2 × 2 × 2 × 3

So, the required LCM = 2 × 2 × 2 × 3 × 3 × 5 = 360

25. The LCM of the three numbers 16, 28 and 42 is:

(a) 168 (b) 2
(c) 252 (d) 336

RRB ALP & Tec. (14-08-18 Shift-I)

Ans : (d) Finding the LCM by using common division method,

2	16	28	42
2	8	14	21
2	4	7	21
2	2	7	21
3	1	7	21
7	1	7	7
	1	1	1

So, the required LCM = 2 × 2 × 2 × 2 × 3 × 7 = 336

26. The LCM of 48 and 54 is :

(a) 6 × 8 × 9 (b) 48 × 54
(c) 6 × 2 × 9 (d) 6 × 8 × 3

RRB ALP & Tec. (13-08-18 Shift-I)

Ans. (a) Finding the LCM by using prime factorization method,

48 = 3×2×2×2×2
54 = 3×3×3× 2

So, the required LCM = 3×2×2×2×2×3×3 = 6×8×9

Type - 2

27. Find the least number which divided by 16, 24, 36 and 54, leaves remainder 12, 20, 32 and 50 respectively?

(a) 432 (b) 444
(c) 428 (d) 452

RRB RPF SI -11/01/2019 (Shift-III)

Ans : (c) According to the question,

16 – 12 = 4, 24 – 20 = 4
36 – 32 = 4 , 54 – 50 = 4

2	16,	24,	36,	54
2	8,	12,	18,	27
2	4,	6,	9,	27
2	2,	3,	9,	27
3	1,	3,	9,	27
3	1,	1,	3,	9
3	1,	1,	1,	3
	1,	1,	1,	1

So, the required number = (LCM of 16,24,36 and 54)–4
= 2×2×2×2×3×3 = 432 – 4 = 428

28. The largest five digit number, which divided by 5, 6 and 7, gives remainder 2 in each case. What is the number?

(a) 99958 (b) 99972
(c) 99858 (d) 99962

RRB RPF Constable -19/01/2019 (Shift-II)

Ans. (d) : The largest five digit number = 99999

LCM of 5, 6 and 7 = 5×6×7 = 210

So,

```
        476
210)99999
    840
    1599
    1470
     1299
     1260
       39
```

Remainder = 39

Hence, the required number = 99999 – 39 = 99960

But in each condition remainder is 2, so the required number = 99960 + 2 = 99962

29. Find the smallest number that leaves the remainder 13, 41 and 29 on being divided by 20, 48 and 36 respectively.

(a) 727 (b) 720
(c) 713 (d) 187

RRB JE - 30/05/2019 (Shift-I)

Ans : (c) According to the question,

(20 – 13) = (48 – 41) = (36 – 29) = 7

So, the required number = (LCM of 20, 48 and 36) – 7

2	20	48	36
2	10	24	18
2	5	12	9
2	5	6	9
3	5	3	9
3	5	1	3
5	5	1	1
	1	1	1

L.C.M. = 2 × 2 × 2 × 2 × 3 × 3 × 5 = 720 – 7 = 713

30. Find least number which divided by 12, 15, 20 and 54 leaves a remainder 8 in each case?

(a) 540 (b) 556
(c) 532 (d) 548

RRB JE - 02/06/2019 (Shift-II)

Ans. (d) The required number = (LCM of 12, 15, 20 and 54)+8

2	12	15	20	54
2	6	15	10	27
3	3	15	5	27
3	1	5	5	9
3	1	5	5	3
5	1	5	5	1
	1	1	1	1

(2 × 2 × 3 × 3 × 3 × 5) + 8
= 540 + 8 = 548

31. What is the smallest number which divided by 6, 7, 8, 9, and 12, gives remainder 2 in each case?

(a) 508 (b) 608
(c) 502 (d) 506

RRB Group-D – 24/10/2018 (Shift-I)

Ans : (d) LCM of 6, 7, 8, 9, and 12.

2	6,7,8,9,12
2	3,7,4,9,6
2	3,7,2,9,3
3	3,7,1,9,3
3	1,7,1,3,1
7	1,7,1,1,1
	1,1,1,1,1

LCM = 2 × 2 × 2 × 3 × 3 × 7 = 504
gives remainder 2 in each case
So, the required number = 504 + 2 = 506

32. What is the smallest number which divided by 4, 6, 10 and 15, gives remainder 3 in each case?
(a) 58 (b) 126
(c) 37 (d) 63

RRB Group-D – 02/11/2018 (Shift-II)

Ans. (d) LCM of 4, 6, 10 and 15.

2	4,6,10,15
2	2,3,5,15
3	1,3,5,15
5	1,1,5,5
	1,1,1,1

LCM = 2 × 2 × 3 × 5 = 60
So, the required number = 60 + 3 = 63

33. What is the greatest number smaller than 5000 which divided by 5, 6 and 7 respectively, leaves remainders of 4, 5 and 6?
(a) 4830 (b) 4829
(c) 4845 (d) 4831

RRB Group-D – 23/10/2018 (Shift-I)

Ans. (b) : According to the question,
Remainder = 5 – 4 = 1, 6 – 5 = 1, 7 – 6 = 1
LCM of 5, 6 and 7 = $5\times6\times7 = 210$
So,

$$\begin{array}{r} 23 \\ 210\overline{)5000} \\ 420 \\ \hline 800 \\ 630 \\ \hline 170 \end{array}$$

Hence, the required number = 5000 – 170 – 1 = 4829

34. What is the smallest number greater than 3000 which divided by 4, 7 and 10 respectively, leaves remainders of 3, 6 and 9?
(a) 3079 (b) 3080
(c) 3081 (d) 3101

RRB Group-D – 23/10/2018 (Shift-II)

Ans. (a) : According to the question,
Remainder = 4 – 3 = 1, 7 – 6 = 1, 10 – 9 = 1
LCM of 4, 7 and 10 = 2 × 2 × 5 × 7 = 140
Let the number = 140 k -1
On, putting k = 22,
The required number = 140 × 22 – 1 = 3079

35. There is a group of 8 children and a dozen packets of pencils for distribution. What is the minimum number of packets to be distributed for each child to get the equal number of pencils?
(a) 4 (b) 3
(c) 2 (d) 1

RRB NTPC 31.03.2016 Shift : 1

Ans : (c) According to the question,
LCM of 12 and 8 = $2\times2\times2\times3 = 24$
So, the number of pencils = 24.
Hence, the least number of packets $= \frac{24}{12} = 2$

36. Find 150% of X, if X is the least number which divided by 6, 7, 8, 9 and 12 leaves remainders 2, 3, 4, 5 and 8 respectively.
(a) 750 (b) 500
(c) 1000 (d) 1200

RRB NTPC 18.04.2016 Shift : 1

Ans : (a) ∵ The number when divided by 6, 7, 8, 9 and 12 leaves remainders 2, 3, 4, 5 and 8 respectively.

$$6 - 2 = 4$$
$$7 - 3 = 4$$
$$8 - 4 = 4$$
$$9 - 5 = 4$$
$$12 - 8 = 4$$

The number = (LCM of 6, 7, 8, 9 and 12) – 4

2	6	7	8	9	12
2	3	7	4	9	6
2	3	7	2	9	3
3	3	7	1	9	3
3	1	7	1	3	1
7	1	7	1	1	1
	1	1	1	1	1

$= (2\times2\times2\times3\times3\times7) - 4 = 504 - 4 = 500$

So, 150% of the number $= 500\times\frac{150}{100}$

$= 5\times150 = 750$

37. Find the smallest number which divided by 12, 15, 18 and 27, leaves the remainders 10, 13, 16 and 25 respectively.
(a) 540 (b) 538
(c) 542 (d) 552

RRB NTPC 16.04.2016 Shift : 1

Ans : (b) According to the question,
12 – 10 = 2
15 – 13 = 2
18 – 16 = 2
27 – 25 = 2
Hence, the required number
= (LCM of 12, 15, 18 and 27) – 2

$12 = 2\times2\times3$
$15 = 3\times 5$
$18 = 2\times3\times3$
$27 = 3\times3\times3$
LCM of 12, 15, 18 and 27 = $2\times2\times3\times3\times3\times5 = 540$
$= 540 - 2 = 538$

38. Find the least number which divided by 11, 16, 21, 25 and 28, leaves remainder 3 in each case.
(a) 92400 (b) 92597
(c) 92403 (d) 92000

RRB NTPC 22.04.2016 Shift : 3

Ans : (c) The required number = (LCM of 11, 16, 21, 25 and 28) + 3

2	11	16	21	25	28
2	11	8	21	25	14
7	11	4	21	25	7
	11	4	3	25	1

$= 2 \times 2 \times 7 \times 11 \times 4 \times 3 \times 25 = 92400$
So, the required number = 92400 + 3 = 92403

39. A natural number when divides by 5, 6, 7 or 8, leaves a remainder of 4 in each case. What is the smallest of all such numbers?
(a) 214 (b) 424
(c) 844 (d) 1264

RRB ALP & Tec. (21-08-18 Shift-III)

Ans : (c) LCM of 5, 6, 7, and 8 = $5\times7\times2\times3\times4 = 840$
According to the question,
The required least number = 840 + 4=844

40. A natural number, when divides by 4, 5, 6 and 7, leaves a remainder of 3 in each case. What is the smallest of all such numbers?
(a) 63 (b) 423
(c) 843 (d) 213

RRB ALP & Tec. (14-08-18 Shift-II)

Ans : (b)
The required least number = (LCM of 4, 5, 6 and 7)+3
= 420+3
= 423

41. Find the least number which divided by 20, 25, 35 and 40 leaves remainder 14, 19, 29 and 34 respectively?
(a) 1364 (b) 1394
(c) 1384 (d) 1374

RRB ALP & Tec. (14-08-18 Shift-III)

Ans : (b) According to the question,
$20 - 14 = 6$
$25 - 19 = 6$
$35 - 29 = 6$
$40 - 34 = 6$
Hence, the required number
= (LCM of 20, 25, 35 and 40) – 6

2	20	25	35	40
2	10	25	35	20
2	5	25	35	10
5	5	25	35	5
5	1	5	7	1
7	1	1	7	1
	1	1	1	1

$= (2\times2\times2\times5\times5\times7) - 6 = 1400 - 6 = 1394$

42. What is the smallest number which, divided by 15 and 20, leaves remainder 9 in each case?
(a) 60 (b) 65
(c) 69 (d) 309

RRB ALP & Tec. (28.08.2015, SHIFT I)

Ans :(c) LCM of 15 and 20
$\because$ $15 = 3\times\boxed{5}$
$20 = 4\times\boxed{5}$
LCM $= 3\times4\times5 = 60$
So, the required number = 60 + 9 = 69

Type - 3

43. What is the HCF of 45, 240 and 315?
(a) 3 (b) 15
(c) 45 (d) 5

RRB RPF SI -16/01/2019 (Shift-I)

Ans. (b) : Multiples of given numbers,
$\because$ $45 = 3\times3\times5 = 3^2\times5$
$240 = 2\times2\times2\times2\times3\times5 = 2^4\times3\times5$
$315 = 3\times3\times5\times7 = 3^2\times5\times7$
So, HCF = 3 × 5 = 15

44. The product of two co-prime numbers is 117. Find their HCF.
(a) 117 (b) 7
(c) 1 (d) 13

RRB JE - 25/05/2019 (Shift-I)

Ans : (c) Co-prime numbers Pair of such numbers whose HCF is 1 are called co-prime numbers.
$117 = 13\times9$
HCF (13, 9) = 1

45. Find the HCF of $2\times3^2\times5^2$, $5\times3\times2^2$ and $5^2\times3\times2^2$.
(a) 150 (b) 30
(c) 60 (d) 90

RRB JE - 25/05/2019 (Shift-II)

Ans : (b) On finding the HCF,
$2 \times 3^2 \times 5^2$
$5 \times 3 \times 2^2$
$5^2 \times 3 \times 2^2$
HCF = $2\times 3 \times 5 = 30$

46. Find the HCF of $a^3b^3c^3$, $a^2b^2c^2$, abc and a^2bc.
(a) $a^4b^4c^4$ (b) $a^3b^3c^3$
(c) $a^2b^2c^2$ (d) abc

RRB JE - 27/06/2019 (Shift-III)

Ans : (d) On finding the HCF,

$a^3b^3c^3 = a \times a \times a \times b \times b \times b \times c \times c \times c$

$a^2b^2c^2 = a \times a \times b \times b \times c \times c$

$abc = a \times b \times c$

$a^2bc = a \times a \times b \times c$

HCF $= a \times b \times c$

Note- For finding the HCF of any expression, always take the least power of numbers.

47. Find the HCF of 3341 and 3328.

(a) 257 (b) 337
(c) 13 (d) 31

RRB RPF SI -13/01/2019 (Shift-III)

Ans : (c) On doing factorization,

$3341 = 13 \times 257$

$3328 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 13$

So, the required HCF = 13

48. What is the HCF of 148 and 370?

(a) 148 (b) 37
(c) 74 (d) 2

RRB RPF Constable -20/01/2019 (Shift-I)

Ans. (c) On finding HCF of 148 and 370,

$148 = 74 \times 2$

$370 = 74 \times 5$

Hence, the required HCF = 74

49. What is the HCF of 343, 217 and 455?

(a) 9 (b) 7
(c) 11 (d) 13

RRB Group-D – 27/11/2018 (Shift-III)

Ans. (b) $343 = 7\times7\times7$

$217 = 7\times31$

$455 = 5\times7\times13$

H.C.F. = 7

50. What is the HCF of 140 and 196?

(a) 14 (b) 28
(c) 21 (d) 42

RRB Group-D – 06/12/2018 (Shift-III)

Ans. (b) : Finding the HCF by factorization method,

$140 = 2 \times 2 \times 5 \times 7$

$196 = 2 \times 2 \times 7 \times 7$

So, the required HCF $= 2 \times 2 \times 7 = 28$

51. The HCF of 64, 28 and 96 is:

(a) 1 (b) 8
(c) 4 (d) 2

RRB Group-D – 20/09/2018 (Shift-III)

Ans : (c) HCF of 64, 28 and 96,

$64 = 2\times2\times2\times2\times2\times2$

$28 = 2\times2\times7$

$96 = 2\times2\times2\times2\times2\times3$

So, the required HCF $= 2\times2 = 4$

52. The value of half of the HCF of 36 and 144 is:

(a) 144 (b) 18
(c) 36 (d) 72

RRB Group-D – 17/09/2018 (Shift-III)

Ans. (b) : On finding the HCF,

$36 = 2 \times 2 \times 3 \times 3$

$144 = 2 \times 2 \times 2 \times 2 \times 3 \times 3$

So, the HCF $= 2 \times 2 \times 3 \times 3 = 36$

So, the required value $= \frac{36}{2} = 18$

53. Find the HCF of 306, 204 and 136.

(a) 54 (b) 68
(c) 34 (d) 17

RRB Group-D – 20/09/2018 (Shift-I)

Ans. (c) : On finding the HCF of 306, 204 and 136,

$306 = 2 \times 3 \times 3 \times 17$

$204 = 2 \times 2 \times 3 \times 17$

$136 = 2 \times 2 \times 2 \times 17$

So, the required HCF $= 2 \times 17 = 34$

54. What is the HCF of 30, 42 and 96?

(a) 8 (b) 6
(c) 4 (d) 2

RRB Group-D – 05/10/2018 (Shift-I)

Ans. (b) : On finding HCF of 30, 42 and 96 by division method,

```
30)42(1
   30
   12)30(2
      24
       6)12(2
         12
         x x
6)96(16
  6
  36
  36
  x x
```

So, the required HCF = 6

55. What is the HCF of 44, 88 and 154?

(a) 8 (b) 11
(c) 44 (d) 22

RRB Group-D – 05/10/2018 (Shift-I)

Ans. (d) : On finding the HCF of given numbers,

$44 = 2 \times 2 \times 11$

$88 = 2 \times 2 \times 2 \times 11$

$154 = 2 \times 11 \times 7$

So, the required HCF $= 2\times11 = 22$

56. Find the HCF of $(5^3 \times 4^3)$, $(3^5 \times 5^2 \times 4^4)$ and $(3^2 \times 5 \times 4^3)$.

(a) 340 (b) 328
(c) 230 (d) 320

RRB Group-D – 16/11/2018 (Shift-I)

Ans. (d) HCF of given expression,

$(5^3 \times 4^3)$, $(3^5 \times 5^2 \times 4^4)$ and $(3^2 \times 5 \times 4^3)$

HCF $= 5 \times 4^3$

$= 5 \times 64 = 320$

57. Find the HCF of 2349, 2835 and 3078.

(a) 81 (b) 9
(c) 27 (d) 3

RRB Group-D – 22/10/2018 (Shift-III)

Ans : (a) On finding HCF by division method,

```
2835) 3078 (1
      2835
       243) 2835 (11
            243
             405
             243
             162) 243 (1
                  162
                   81) 162 (2
                       162
                        ××
81) 2349 (29
    162
     729
     729
     ×××
```

So, it is clear that the required HCF is 81.

58. $59 \times 29 = 1711$, then what is the HCF of 6844 and 354?

(a) 236 (b) 118
(c) 177 (d) 59

RRB Group-D – 15/11/2018 (Shift-III)

Ans. (b) : On finding HCF by division method,

```
354 ) 6844 (19
      354
      ----
      3304
      3186
       118 ) 354 (3
             354
             ---
             XXX
```

So, the required HCF = 118

59. Find the HCF of $(3^3 \times 5^3 \times 6^3)$, $(3^2 \times 3^5 \times 5^2 \times 6^4)$, $(3^3 \times 3^2 \times 5 \times 6^3)$

(a) 1560 (b) 1600
(c) 1280 (d) 29160

RRB Group-D – 15/11/2018 (Shift-III)

Ans : (d) : From given expressions,

$3^3 \times 5^3 \times 6^3 \Rightarrow 3^3 \times 5^3 \times 6^3$

$3^2 \times 3^5 \times 5^2 \times 6^4 \Rightarrow 3^7 \times 5^2 \times 6^4$

$3^3 \times 3^2 \times 5 \times 6^3 \Rightarrow 3^5 \times 5 \times 6^3$

So, the required HCF = $3^3 \times 5 \times 6^3 = 27 \times 5 \times 216$
$= 29160$

60. 245 and 343 guests will come on Sohrab's birthday. Chocolate is to be offered to each guest. The seller said that she would pack the chocolate in the carton and take back the unopened carton, but she will add the packing cost for the carton she has to pack. In this situation, how much chocolate should Sohrab hold in each carton?

(a) 77 (b) 7
(c) 21 (d) 49

RRB Group-D – 12/11/2018 (Shift-I)

Ans. (d) :
The number of the chocolates in each carton = HCF of 245 and 343
So, on finding the HCF by division method,

```
245) 343 ( 1
     245
      98) 245 ( 2
          196
           49) 98 ( 2
               98
               XX
```

So, the required HCF is 49.

61. The number of students in a school in 4th, 5th and 6th grade was 188, 282 and 423. If the class is divided into sections and each section had the same number of students, so what was the total minimum number of sections of these three classes?

(a) 20 (b) 18
(c) 19 (d) 17

RRB Group-D – 12/11/2018 (Shift-III)

Ans : (c) On finding the HCF of 188, 282 and 423 by division method,

```
188 ) 282 ( 1        94) 423 ( 4
      188                376
       94) 188 ( 2       47) 94 ( 2
           188              94
           XXX              XX
```

So, HCF = 47
So, the number of students in each section of each class = 47

$\frac{188}{47} = 4, \quad \frac{282}{47} = 6, \quad \frac{423}{47} = 9$

Hence, the total minimum number of sections of three classes = 4 + 6 + 9 = 19

62. What is the HCF of 40, 36, 15 and 24?

(a) 1 (b) 5
(c) 3 (d) 2

RRB Group-D – 24/10/2018 (Shift-III)

Ans. (a) : On finding HCF by factorization method,

$40 = 2 \times 2 \times 2 \times 5$
$36 = 2 \times 2 \times 3 \times 3$
$15 = 3 \times 5$
$24 = 2 \times 2 \times 2 \times 3$

So, the required HCF = 1

63. What is the HCF of 156, 312 and 195?

(a) 78 (b) 3
(c) 13 (d) 39

RRB Group-D – 11/10/2018 (Shift-I)

Ans : (d) On finding HCF by factorization method,

$156 = 2 \times 2 \times 3 \times 13$
$312 = 2 \times 2 \times 2 \times 3 \times 13$
$195 = 3 \times 5 \times 13$

So, the required HCF = $3 \times 13 = 39$

64. If HCF of $2x^2 + 5x - 12$ and $x^2 + x - 12$ is (x+a), then find the value of a.

(a) –3 (b) –2
(c) 4 (d) 5

RRB NTPC 16.04.2016 Shift : 1

Ans : (c) First term,

$2x^2 + 5x - 12$
$= 2x^2 - 3x + 8x - 12$
$= x(2x-3) + 4(2x-3)$
$= (x+4)(2x-3)$

Second term,

$= x^2 + x - 12$
$= x^2 + 4x - 3x - 12$
$= x(x+4) - 3(x+4)$
$= (x+4)(x-3)$

So, from both terms the HCF = (x + 4)

And, on comparison $(x+4)$ with $(x+a)$, $\Rightarrow \boxed{a = 4}$

65. Find the HCF of 1757, 2259.

(a) 231 (b) 241
(c) 251 (d) 261

RRB NTPC 11.04.2016 Shift : 1

Ans : (c) On finding the HCF by division method,

1757) 2259 (1
1757
502) 1757 (3
1506
251) 502 (2
502
×××

So, the required HCF = 251.

66. Sheeba has 24 chocolates, 36 biscuits and 60 ice creams to distribute to her classmates. She wants each of her classmates to get the same number of each thing. What is the maximum number of classmates in which she can distribute completely without saving a single thing?

(a) 6 (b) 18
(c) 12 (d) 15

RRB NTPC 29.04.2016 Shift : 1

Ans : (c) According to the question,

The number of chocolates, biscuits and ice creams is 24, 36 and 60 respectively.

So, the number of students = HCF of 24, 36 and 60

$24 = 2\times2\times2\times3$

$36 = 2\times2\times3\times3$

$60 = 2\times2\times3\times5$

HCF = $2\times2\times3 = 12$

67. What is the HCF of 2189 and 2587?

(a) 3 (b) 197
(c) 199 (d) 198

RRB NTPC 30.04.2016 Shift : 3

Ans : (c) On finding the HCF by division method,

2189) 2587 (1
2189
398) 2189 (5
1990
199) 398 (2
398
×××

So, the required HCF is 199.

68. The HCF of 56, 140 and 168 is:

(a) 28 (b) 7
(c) 14 (d) 4

RRB ALP & Tec. (30-08-18 Shift-I)

Ans : (a) On finding the HCF by factorization method,

$56 = 2 \times 2 \times 2 \times 7$
$140 = 2 \times 2 \times 5 \times 7$
$168 = 2 \times 2 \times 2 \times 7 \times 3$

So, the HCF = $2 \times 2 \times 7 = 28$

69. The HCF of 162, 54 and 135 is:

(a) 9 (b) 1
(c) 27 (d) 3

RRB ALP & Tec. (21-08-18 Shift-II)

Ans : (c) On finding the HCF by division method,

54) 135 (2
108
27) 54 (2
54
×

27) 162 (6
162
×

So, the required HCF is 27.

70. The HCF of 24, 60 and 90 is :

(a) 3 (b) 6
(c) 12 (d) 4

RRB ALP & Tec. (20-08-18 Shift-III)

Ans : (b) On finding the HCF by division method,

24)60(2
48
12)24(2
24
×

12)90(7
84
6)12(2
12
×

So, the required HCF is 6.

71. The HCF of 20, 28 and 48 is:

(a) 8 (b) 2
(c) 1 (d) 4

RRB ALP & Tec. (14-08-18 Shift-I)

Ans : (d) On finding the HCF by factorization method,

$20 = 2 \times 2 \times 5$
$28 = 2 \times 2 \times 7$
$48 = 2 \times 2 \times 2 \times 2 \times 3$

So, the required HCF = $2 \times 2 = 4$

72. The HCF of 36, 54 and 108 is:

(a) 6 (b) 9
(c) 18 (d) 12

RRB ALP & Tec. (14-08-18 Shift-II)

Ans : (c) On finding the HCF by factorization method,

$36 = 2\times2\times3\times3$
$54 = 2\times3\times3\times3$
$108 = 2\times2\times3\times3\times3$

So, the required HCF = $2\times3\times3 = 18$

73. What is the HCF of 36, 72 and 126?

(a) 18 (b) 36
(c) 9 (d) 12

RRB ALP & Tec. (13-08-18 Shift-II)

Ans : (a) On finding the HCF by factorization method,

$36 = 2 \times 2 \times 3 \times 3$
$72 = 2 \times 2 \times 2 \times 3 \times 3$
$126 = 2 \times 3 \times 3 \times 7$

So, the required HCF = $2 \times 3 \times 3 = 18$

Type - 1

74. Find the largest possible length that can be used to measure the length of 2m 76cm, 5m 52cm and 11m 96cm.

(a) 92 cm (b) 11.96 cm
(c) 92 m (d) 1196 cm

RRB RPF SI -12/01/2019 (Shift-III)

Ans : (a)

$2m\ 76cm = 2\times100+76 = 276cm$

$5m\ 52cm = 5\times100+52 = 552cm$

$11m\ 96cm = 11\times100+96 = 1196cm$

So, the largest possible length = HCF of 276, 552 and 1196.

552)1196(2
1104
92)552(6
552
×××

92)276(3
276
×××

HCF = 92cm

75. Find the largest four digit number which divided by 8, 12 and 16, gives remainder 2.

(a) 9984 (b) 9985
(c) 9986 (d) 9987

RRB RPF Constable -22/01/2019 (Shift-II)

Ans : (c) LCM of 8, 12 and 16

2	8,	12,	16
2	4,	6,	8
2	2,	3,	4
2	1,	3,	2
3	1,	3,	1
	1,	1,	1

LCM = $2 \times 2 \times 2 \times 2 \times 3 = 48$

The largest 4-digit number = 9999

48)9999(208
96
399
384
×15

So, the largest 4-digit number divisible by 48
= 9999 – 15 = 9984

According to the question, the remainder is 2.

So, the required number = 9984 + 2 = 9986

76. Find the largest number, which divides 1580 and 3800, so that the remainder is 8 and 1 respectively.

(a) 262 (b) 131
(c) 65.5 (d) 393

RRB NTPC 19.04.2016 Shift : 2

Ans : (b) The required number = HCF of (1580 – 8) and (3800 – 1)

So, the HCF of 1572 and 3799

1572)3799(2
3144
655)1572(2
1310
262)655(2
524
131)262(2
262
×××

So, the required number is 131.

77. Find such greatest number which gives equal remainders in each case when divides 76, 151 and 226.

(a) 69 (b) 60
(c) 75 (d) 70

RRB Group-D – 16/10/2018 (Shift-III)

Ans : (c) If the remainder is same, then

$226 - 151 = 75 \times 1$
$151 - 76 = 75 \times 1$
$226 - 76 = 75 \times 2$

So, the required number = HCF of 75, 75 and 150.
= 75

78. What is the largest number which divides 1657 and 2037, gives remainders 6 and 5 respectively?

(a) 150 (b) 125
(c) 127 (d) 130

RRB Group-D – 05/10/2018 (Shift-I)

Ans. (c) : 1657 – 6 = 1651
2037 – 5 = 2032

So, the HCF of 1651 and 2032,

1651)2032(1
1651
381)1651(4
1524
127)381(3
381
×××

So, the required number is 127.

79. Find the largest number which divides 200 and 432, gives remainders 2 and 3 respectively.

(a) 24 (b) 8
(c) 33 (d) 44

RRB Group-D – 26/11/2018 (Shift-III)

Ans : (c) According to the question,

$200 - 2 = 198$

$432 - 3 = 429$

So, the HCF of 198 and 429 is the required number,

$$198\overline{)429}\,(2$$
$$\underline{396}$$
$$33\overline{)198}\,(6$$
$$\underline{198}$$
$$\times\times\times$$

So, the required number is 33.

80. What is the largest number which, divides 63, 77 and 98, gives remainders 3, 5 and 2 respectively?

(a) 10 (b) 9
(c) 6 (d) 8

RRB Group-D – 05/11/2018 (Shift-II)

Ans : (c) According to the question,

$63 - 3 = 60$

$77 - 5 = 72$

$98 - 2 = 96$

So, the required number = HCF of 60, 72 and 96.

$60 = 2\times2\times3\times5$

$72 = 2\times2\times2\times3\times3$

$96 = 2\times2\times2\times2\times2\times3$

HCF = $2\times2\times3 = 12$

While 12 is not in the option but 12 will be divisible by 6.

81. Find the greatest length which can be used to measure exactly three cloth pieces of length 1.26 m, 1.98 m and 1.62 m respectively.

(a) 12 cm (b) 14 cm
(c) 16 cm (d) 18 cm

RRB NTPC 31.03.2016 Shift : 2

Ans : (d) The required length = HCF of 126cm, 198cm and 162 cm.

$$126\overline{)198}\,(1 \qquad 18\overline{)162}\,(9$$
$$\underline{126} \qquad\qquad \underline{162}$$
$$72\overline{)126}\,(1 \qquad\qquad \times\times\times$$
$$\underline{72}$$
$$54\overline{)72}\,(1$$
$$\underline{54}$$
$$18\overline{)54}\,(3$$
$$\underline{54}$$
$$\times\times$$

So, the HCF = 18cm.

Hence, the greatest length = 18cm

82. 50 pens, 80 pencils and 65 scales were distributed among some students and found that five out of each item were not distributed. Find the number of students.

(a) 5 (b) 20
(c) 15 (d) 10

RRB NTPC 31.03.2016 Shift : 3

Ans : (c) According to the question,

$50 - 5 = 45$

$80 - 5 = 75$

$65 - 5 = 60$

So, the required number of students = HCF of 45, 75 and 60.

$45 = 3\times3\times5$

$75 = 3\times5\times5$

$60 = 2\times2\times3\times5$

So, HCF = $3\times5 = 15$

83. You have 20 big and 16 small diaries and want to make gift packs containing both in each pack. What is the maximum number of gift packs you can make without any left over?

(a) 5 (b) 4
(c) 3 (d) 2

RRB NTPC 29.03.2016 Shift : 3

Ans : (b) the maximum number of gift packs = HCF of 20 and 16.

$$16\overline{)20}\,(1$$
$$\underline{16}$$
$$4\overline{)16}\,(4$$
$$\underline{16}$$
$$\times\times$$

So, HCF = 4,

So, 4 maximum gift packets can be made.

84. Find such greatest number which gives equal remainders in each case when divides 270, 675 and 1215.

(a) 45 (b) 135
(c) 270 (d) 75

RRB NTPC 19.04.2016 Shift : 3

Ans : (b) According to the question,

$675 - 270 = 405 = 3\times3\times3\times3\times5$

$1215 - 675 = 540 = 2\times2\times3\times3\times3\times5$

$1215 - 270 = 945 = 3\times3\times3\times5\times7$

$HCF = 3\times3\times3\times5 = 135$

So, the required number is 135.

85. Find such greatest number which gives equal remainders in each case when divides 60, 150 and 285.

(a) 30 (b) 25
(c) 45 (d) 55

RRB NTPC 22.04.2016 Shift : 1

Ans : (c) HCF of 60, 150 and 285,

$60 = 2\times2\times3\times5$

$150 = 2\times3\times5\times5$

$285 = 3\times5\times19$

$HCF = 3\times5 = 15$

So, the number will be 15k where k = 1, 2, 3, ……

Take k = 3, The required number = $15\times3 = 45$

$\Rightarrow \frac{60}{45} = 1\,quotient + 15\ remainder$

$\Rightarrow \frac{150}{45} = 3\ quotient\ +\ 15\ remainder$

$\Rightarrow \frac{285}{45} = 6\ quotient\ +\ 15\ remainder$

$\because$ Remainder is same.

So, the required number is 45.

86. Find the greatest number which on dividing 391 and 318 leaves remainders 7 and 6 respectively.

(a) 20 (b) 23
(c) 24 (d) 32

RRB NTPC 27.04.2016 Shift : 2

Ans : (c) According to the question,

$391 - 7 = 384$

$318 - 6 = 312$

The HCF of 384 and 312,

$384 = 2\times2\times2\times2\times2\times2\times2\times3$

$312 = 2\times2\times2\times3\times13$

$\therefore HCF = 2\times2\times2\times3 = 24$

So, the required number is 24.

Type - 1

87. What is the HCF of $\frac{34}{9}, \frac{85}{18}$ and $\frac{68}{45}$.

(a) $\frac{17}{45}$ (b) $\frac{34}{9}$
(c) $\frac{17}{90}$ (d) $\frac{1}{90}$

RRB RPF Constable -25/01/2019 (Shift-I)

Ans : (c) HCF of $\frac{34}{9}, \frac{85}{18}, \frac{68}{45}$

$\frac{34}{9} = \frac{2\times17}{3\times3}$

$\frac{85}{18} = \frac{5\times17}{3\times3\times2}$

$\frac{68}{45} = \frac{2\times2\times17}{3\times3\times5}$

HCF of fractions $= \frac{\text{The HCF of numerator}}{\text{The LCM of denominator}} = \frac{17}{90}$

88. Find the LCM and HCF of 1.75, 5.6 and 7.

(a) 24, 0.2 (b) 28, 0.25
(c) 28, 0.35 (d) 24, 0.35

RRB JE - 23/05/2019 (Shift-I)

Ans : (c)

LCM of 1.75, 5.6 and 7 = LCM of $\frac{175}{100}, \frac{560}{100}, \frac{700}{100}$

LCM $= \frac{\text{The LCM of numerator}}{\text{The HCF of denominator}}$

$= \frac{2800}{100} = 28$

HCF of $\frac{175}{100}, \frac{560}{100}, \frac{700}{100}$

$= \frac{\text{The HCF of numerator}}{\text{The LCM of denominator}} = \frac{35}{100} = 0.35$

Hence, LCM and HCF = 28 and 0.35

89. Find the LCM of 36/225, 48/150 and 72/85.

(a) 144/5 (b) 72/85
(c) 140/1 (d) 150/22

RRB JE - 23/05/2019 (Shift-III)

Ans : (a) LCM of given numbers

$= \frac{\text{The LCM of numerator}}{\text{The HCF of denominator}}$

$= \frac{LCM\ of\ 36,48,72}{HCF\ of\ 225,150,85}$

$= \frac{144}{5}$

90. Find the GCD of 1.08, 0.36 and 0.9.

(a) 0. 03 (b) 18
(c) 0. 18 (d) 1. 8

RRB JE - 24/05/2019 (Shift-I)

Ans : (c) GCD of 1.08, 0.36 and 0.9

$= GCD\,or\,HCF\ of\ \frac{108}{100}, \frac{36}{100}, \frac{90}{100} = \frac{GCD\ of\ 108,36,90}{LCM\ of\ 100,100,100}$

$= \frac{18}{100} = 0.18$

91. Find the LCM of 1.05 and 2.1.

(a) 2.1 (b) 0.6
(c) 1.25 (d) 4

RRB JE - 26/05/2019 (Shift-I)

Ans : (a) From given numbers,

$1.05 = \frac{105}{100} = \frac{21}{20}$

$2.1 = \frac{21}{10}$

LCM of $\frac{21}{20}, \frac{21}{10} = \frac{\text{The LCM of numerator}}{\text{The HCF of denominator}}$

$= \frac{21}{10} = 2.1$

92. Find the LCM of 2/3 and 6/7.

(a) 6 (b) 6/21
(c) 12/21 (d) 2/21

RRB RPF SI -10/01/2019 (Shift-I)

Ans : (a)

The LCM of $\frac{2}{3}, \frac{6}{7} = \frac{\text{The LCM of numerator}}{\text{The HCF of denominator}}$

$= \frac{6}{1} = 6$

93. Find the HCF of 36/75, 48/50 and 72/30.

(a) 144/50 (b) 144/45
(c) 12/75 (d) 12/150

RRB JE - 27/05/2019 (Shift-III)

Ans : (d)

The HCF of $\frac{36}{75},\frac{48}{50},\frac{72}{30} = \frac{\text{The HCF of numerator}}{\text{The LCM of denominator}}$

$= \frac{HCF\ of\ 36,48,72}{LCM\ of\ 75,50,30} = \frac{12}{150}$

94. LCM of a set of fractions is:

(a) LCM of numerators/ HCF of denominators
(b) HCF of numerators/ LCM of denominators
(c) LCM of numerators/ LCM of denominators
(d) HCF of numerators/ HCF of denominators

RRB JE - 30/05/2019 (Shift-I)

Ans : (a)

LCM of a fraction = $\frac{\text{The LCM of numerator}}{\text{The HCF of denominator}}$

95. Find the HCF of 1.43, 1.87 and 20.9.

(a) 0.011 (b) 0.10
(c) 1.1 (d) 0.11

RRB NTPC 12.04.2016 Shift : 2

Ans : (d) From given numbers,

$1.43 = \frac{143}{100}$

$1.87 = \frac{187}{100}$

$20.9 = \frac{2090}{100}$

So, HCF of numbers = $\frac{\text{The HCF of numerator}}{\text{The LCM of denominator}}$

$\text{HCF} = \frac{11}{100} = 0.11$

96. Find the HCF of 0.32, 2.72, 12.8 and 14.4.

(a) 16 (b) 1.6
(c) 0.16 (d) 2.72

RRB NTPC 16.04.2016 Shift : 2

Ans : (c) From given numbers,

$0.32 = \frac{32}{100}$, $12.8 = \frac{128}{10}$

$2.72 = \frac{272}{100}$, $14.4 = \frac{144}{10}$

So, the HCF of given numbers =

$\frac{\text{The HCF of numerator}}{\text{The LCM of denominator}}$

$= \frac{HCF\ of\ 32,272,128,144}{LCM\ of\ 100,100,10,10} = \frac{16}{100} = 0.16$

97. The HCF of $\frac{2}{3},\frac{8}{9},\frac{10}{27}$ and $\frac{32}{81}$ is:

(a) $\frac{160}{3}$ (b) $\frac{160}{81}$
(c) $\frac{2}{3}$ (d) $\frac{2}{81}$

RRB Group-D – 17/09/2018 (Shift-I)

Ans : (d) HCF of given numbers,

$= \frac{\text{The HCF of numerator}}{\text{The LCM of denominator}}$

$= \frac{2}{3},\frac{8}{9},\frac{10}{27},\frac{32}{81}$

$= \frac{2}{81}$

98. What is the HCF of $\frac{3}{5},\frac{9}{10}$ and $\frac{6}{25}$?

(a) $\frac{3}{50}$ (b) $\frac{3}{5}$
(c) 18 (d) $\frac{18}{50}$

RRB Group-D – 29/10/2018 (Shift-III)

Ans : (a) The HCF of given fractions,

$\frac{3}{5},\frac{9}{10}\ and\ \frac{6}{25}$

$= \frac{\text{The HCF of numerator}}{\text{The LCM of denominator}}$

$= \frac{HCF\ of\ 3,9,6}{LCM\ of\ 5,10,25} = \frac{3}{50}$

99. Find the LCM of 0.63, 10.5, 2.1 and 4.20.

(a) 63 (b) 0.63
(c) 6.30 (d) 6300

RRB NTPC 16.04.2016 Shift : 1

Ans : (a) According to the question,

$0.63 = \frac{63}{100}, 10.5 = \frac{105}{10}, 2.1 = \frac{21}{10}, 4.20 = \frac{420}{100} = \frac{42}{10}$

So, the LCM of $\frac{63}{100},\frac{105}{10},\frac{21}{10},\frac{42}{10}$

$= \frac{LCM\ of\ 63,105,21,42}{HCF\ of\ 100,10,10,10}$

$= \frac{21\times3\times5\times2}{10} = \frac{630}{10} = 63$

100. Find the LCM of $\frac{1}{2},\frac{2}{5},\frac{4}{7}$ and $\frac{6}{17}$.

(a) 6 (b) 18
(c) 24 (d) 12

RRB Group-D – 25/09/2018 (Shift-I)

Ans : (d) LCM of fractions

$= \frac{\text{The HCF of numerator}}{\text{The LCM of denominator}}$

$= \frac{LCM\ of\ 1,2,4,6}{HCF\ of\ 2,5,7,17} = \frac{12}{1} = 12$

101. Find the HCF of $\frac{7}{9},\frac{14}{15},\frac{7}{10}$.

(a) $\frac{7}{90}$ (b) $\frac{7}{675}$
(c) $\frac{7}{45}$ (d) $\frac{14}{45}$

RRB Group-D – 05/11/2018 (Shift-III)

Ans. (a) HCF of given fractions

$$= \frac{\text{The HCF of numerator}}{\text{The LCM of denominator}}$$

$$\frac{\text{HCF of } 7,14,7}{\text{LCM of } 9,15,10} = \frac{7}{90}$$

102. What is the HCF of $\frac{15}{14}, \frac{12}{35}$ and $\frac{40}{49}$.

(a) $\frac{1}{7}$ (b) $\frac{3}{490}$

(c) $\frac{1}{490}$ (d) $\frac{1}{49}$

RRB Group-D – 25/10/2018 (Shift-II)

Ans : (c) The HCF of given fractions,

$$= \frac{\text{HCF of } 15,12,40}{\text{LCM of } 14,35,49}$$

So, HCF of 15, 12 and 40,

$15 = 3\times5$

$12 = 2\times2\times3$

$40 = 2\times2\times2\times5$

$HCF = 1$

And LCM of 14, 35 and 49,

$14 = 2\times7$

$35 = 5\times7$

$49 = 7\times7$

$LCM = 7\times2\times5\times7 = 490$

So, the required HCF = $\frac{1}{490}$

103. What is the LCM of $\frac{9}{8}, \frac{15}{16}$ and $\frac{27}{40}$?

(a) 135 (b) $\frac{135}{80}$

(c) $16\frac{7}{8}$ (d) $33\frac{3}{4}$

RRB Group-D – 23/10/2018 (Shift-III)

Ans : (c) LCM of 9, 15 and 27 = 135

HCF of 8, 16 and 40 = 8

So, the required LCM = $\frac{\text{The LCM of numerator}}{\text{The HCF of denominator}}$

$$= \frac{135}{8} = 16\frac{7}{8}$$

104. Find the LCM of 17/31, 34/62 and 48/93.

(a) 816/31 (b) 802/31

(c) 912/31 (d) 804/31

RRB NTPC 04.04.2016 Shift : 3

Ans : (a) The LCM of given fractions,

LCM of 17, 34 and 48,

$17 = 1\times17$

$34 = 1\times2\times17$

$48 = 1\times2\times2\times2\times2\times3$

$LCM = 1\times2\times2\times2\times2\times3\times17 = 816$

HCF of 31, 62 and 93,

$31 = 1\times31$

$62 = 1\times2\times31$

$93 = 1\times3\times31$

$HCF = 1\times31 = 31$

So, the required LCM = $\frac{\text{The LCM of numerator}}{\text{The HCF of denominator}}$

$$= \frac{816}{31}$$

105. Find the LCM of the following.
2/3, 8/9, 16/27, 32/81

(a) 32/81 (b) 81/32

(c) 32/3 (d) 11/41

RRB NTPC 12.04.2016 Shift : 3

Ans : (c) According to the question,

LCM of $\frac{2}{3}, \frac{8}{9}, \frac{16}{27}, \frac{32}{81} = \frac{\text{LCM of } 2,8,16 \text{ and } 32}{\text{HCF of } 3,9,27 \text{ and } 81}$

$$= \frac{32}{3}$$

106. Find the LCM of $\frac{2}{30}, \frac{20}{40}, \frac{4}{50}, \frac{8}{60}$.

(a) $\frac{40}{300}$ (b) $\frac{2}{10}$

(c) $\frac{1}{4}$ (d) 4

RRB NTPC 11.04.2016 Shift : 3

Ans : (d) LCM of the fractions

$$= \frac{\text{The LCM of numerator}}{\text{The HCF of denominator}}$$

$$= \frac{\text{LCM of } 2,20,4,8}{\text{HCF of } 30,40,50,60} = \frac{40}{10} = 4$$

107. Find the LCM of 13/31, 23/62 and 48/93.

(a) 14352/31 (b) 14452/31

(c) 15432/31 (d) 12534/31

RRB NTPC 10.04.2016 Shift : 3

Ans : (a) LCM of the fractions

$$= \frac{\text{The HCF of numerator}}{\text{The LCM of denominator}}$$

$$= \frac{LCM\ of\ 13,23,48}{HCF\ of\ 31,62,93} = \frac{14352}{31}$$

108. Find the LCM of $\frac{4}{5}, \frac{2}{3}$ and $\frac{5}{7}$

(a) 25 (b) 20

(c) 40 (d) 30

RRB ALP & Tec. (09-08-18 Shift-III)

Ans : (b) LCM of the fractions

$$= \frac{\text{The LCM of numerator}}{\text{The HCF of denominator}}$$

$$= \frac{LCM\ of\ 4,2,5}{HCF\ of\ 5,3,7} = \frac{20}{1} = 20$$

Type - 1

109. Find the HCF and the LCM of 570 and 1425.

(a) 285, 2750 (b) 285, 2850
(c) 289, 2650 (d) 185, 2850

RRB RPF Constable -19/01/2019 (Shift-II)

Ans : (b) HCF of given numbers,

$570 = 2\times3\times5\times19$

$1425 = 3\times5\times5\times19$

$HCF = 3\times5\times19 = 285$

LCM of given numbers,

2	570	1425
3	285	1425
5	95	475
5	19	95
19	19	19
	1	1

$LCM = 2\times3\times5\times5\times19 = 2850$

110. The HCF and LCM of two numbers are 12 and 720 respectively. How many possible pairs are of these numbers?

(a) 3 (b) 4
(c) 2 (d) 1

RRB RPF SI -05/01/2019 (Shift-I)

Ans. (b) : Product of co-factors $= \frac{\text{LCM}}{\text{HCF}}$

$$= \frac{720}{12} = 60$$

Possible pairs - (1, 60), (4, 15), (3, 20), (5, 12)
So, the number of possible pairs is 4.

111. Find the LCM and HCF of inverse of 18 and 24.

(a) 1/6., 1/4 (b) 72.6
(c) 1/72. 1/6 (d) 1/6.1/7

RRB JE - 22/05/2019 (Shift-I)

Ans : (c) Inverse of $18 = \frac{1}{18}$

Inverse of $24 = \frac{1}{24}$

LCM of $\frac{1}{18}, \frac{1}{24} = \frac{\text{LCM of } 1,1}{\text{HCF of } 18,24} = \frac{1}{6}$

HCF of $\frac{1}{18}, \frac{1}{24} = \frac{\text{HCF of } 1,1}{\text{LCM of } 18,24} = \frac{1}{72}$

112. Two numbers are in a ratio 3:5. Their LCM is 75. Find their HCF.

(a) 3 (b) 5
(c) 4 (d) 15

RRB JE - 22/05/2019 (Shift-III)

Ans : (b) Let the numbers be 3x and 5x.
So, LCM of 3x and 5x = 15x
According to the question,

$15x = 75$

$x = \frac{75}{15}$

$x = 5$

So, both the numbers will be,

$3x = 3 \times 5 = 15$

$5x = 5 \times 5 = 25$

So, the HCF of 25 and 15 = 5

113. The LCM and the HCF of two numbers are 693 and 11 respectively. If one is 99, find the other number.

(a) 77 (b) 79
(c) 12 (d) 34

RRB JE - 26/05/2019 (Shift-III)

Ans : (a)
LCM × HCF = First number × Second number

693 × 11 = 99 × Second number

So, the second number $= \frac{693\times11}{99} = 77$

114. Three numbers in the ratio 3:4:5 have the LCM 2400. What is the HCF?

(a) 80 (b) 40
(c) 120 (d) 200

RRB JE - 27/05/2019 (Shift-I)

Ans : (b) Let the numbers are 3x, 4x and 5x.
So, the LCM of 3x, 4x and 5x = 60x

$\Rightarrow$ 60x = 2400

$\Rightarrow$ x = 40

So, the numbers = 120, 160, 200
HCF of 120, 160, 200

$120 \Rightarrow \underline{2 \times 2 \times 2} \times 3 \times \underline{5}$

$160 \Rightarrow \underline{2 \times 2 \times 2} \times 2 \times 2 \times \underline{5}$

$200 \Rightarrow \underline{2 \times 2 \times 2} \times 5 \times \underline{5}$

Hence, HCF = 2 × 2 × 2 × 5 = $\boxed{40}$

115. The LCM and the HCF of two numbers are 2400 and 16 respectively. If one is 480, find the other number.

(a) 40 (b) 60
(c) 90 (d) 80

RRB RPF SI -06/01/2019 (Shift-III)

Ans : (d) According to the question,

LCM = 2400
HCF = 16

First number = 480

$\because$ LCM × HCF = First number × Second number

2400 × 16 = 480 × Second number

Second number $= \frac{2400\times16}{480} = 80$

116. The product of two numbers is 2772 and their LCM is 462. Find the HCF of these numbers.

(a) 18 (b) 21
(c) 6 (d) 42

RRB RPF Constable -19/01/2019 (Shift-III)

Ans : (c) Product of two numbers = LCM × HCF

$2772 = 462 \times HCF$

$HCF = \frac{2772}{462}$

$= 6$

117. The HCF of two numbers in the ratio 15:11 is 13. Find their LCM.

(a) 4290 (b) 2145
(c) 27885 (d) 165

RRB JE - 02/06/2019 (Shift-III)

Ans : (b) Let the numbers are 15x and 11x.

HCF = x

x = 13

So, the numbers are $15 \times 13 = 195$, $11 \times 13 = 143$.

So, the LCM of 195 and 143,

$195 = 3 \times 5 \times 13$

$143 = 11 \times 13$

$LCM = 3 \times 5 \times 11 \times 13 = 2145$

118. The sum of the LCM and the HCF of two numbers is 372. If LCM is equal to 92 times of HCF and one number is 368, find the other number.

(a) 360 (b) 4
(c) 92 (d) 96

RRB JE - 01/06/2019 (Shift-III)

Ans. (b) Let HCF = x

Then LCM = 92x

LCM + HCF = 372

92 x + x = 372

$x = \frac{372}{93}$

x = 4

LCM × HCF = First number × Second number

$92x \times x = 368 \times$ Second number

Second number = $\frac{92 \times x \times x}{368} = \frac{92 \times 4 \times 4}{368} = 4$

119. The product of two numbers is 3192 and their LCM is 56, then find their HCF.

(a) 58 (b) 59
(c) 56 (d) 57

RRB RPF Constable -17/01/2019 (Shift-I)

Ans : (d) Product of the numbers = 3192

LCM = 56

So, HCF = $\frac{Product\ of\ numbers}{LCM}$

$= \frac{3192}{56} = 57$

120. The LCM of two positive integers x and y (x > y) is xy. What is the HCF of both the numbers?

(a) 1
(b) x + y
(c) Cannot be determined
(d) x – y

RRB Group-D – 03/12/2018 (Shift-III)

Ans. (a) : LCM × HCF = First number × Second number

xy × HCF = x × y

\ HCF = 1

121. The LCM of two numbers is 2604 and HCF is 12. If one from them is 84, then find the other number.

(a) 327 (b) 273
(c) 372 (d) 237

RRB Group-D – 25/09/2018 (Shift-III)

Ans. (c) : According to the question,

LCM = 2604 and HCF = 12

First number = 84

Let the other number is x.

Formula–

LCM × HCF = First number × Second number

$2604 \times 12 = 84 \times x$

$x = \frac{2604 \times 12}{84} = \frac{2604}{7} = 372$

Hence, the other required number is 372.

122. The HCF and LCM of two numbers are 29 and 1015 respectively. If one from them is 145, then what is the other number?

(a) 203 (b) 319
(c) 361 (d) 377

RRB Group-D – 05/12/2018 (Shift-III)

Ans : (a) Let the other number is x.

LCM × HCF = First number × Second number

$29 \times 1015 = 145 \times x$

$x = \frac{29 \times 1015}{145} = \frac{29435}{145} = 203$

Hence, Second number = 203

123. The LCM of two numbers is 2268. If their HCF is 12 and one of them is 324 then, what is the other number?

(a) 189 (b) 84
(c) 7 (d) 27

RRB Group-D – 27/11/2018 (Shift-III)

Ans. (b) Let the other number is x.

LCM × HCF = First number × Second number

x × 324 = 2268 × 12

$x = \frac{2268 \times 12}{324}$

$x = \frac{27216}{324}$

x = 84

So, the other required number is 84.

124. The HCF of 560 and one other number is 70 and their LCM is 2800. The other number is:
(a) 840 (b) 350
(c) 700 (d) 280
RRB Group-D – 16/11/2018 (Shift-III)

Ans. (b) :
First number × Second number = HCF × LCM
560 × Second number = 70 × 2800
Second number = $\frac{70 \times 2800}{560}$
$= \frac{2800}{8} = 350$

125. The HCF of 296 and other number is 74 and their LCM is 2072. The other number is:
(a) 1036 (b) 518
(c) 592 (d) 148
RRB Group-D – 15/11/2018 (Shift-I)

Ans : (b) As we know,
First number × Second number = LCM × HCF
296 × Second number = 2072 × 74
Second number = $\frac{2072 \times 74}{296}$
$= \frac{2072}{4} = 518$

126. The LCM of two numbers is 42 times their HCF. The sum of LCM and HCF is 602. If one from them is 84, then find the other number.
(a) 98 (b) 78
(c) 87 (d) 89
RRB Group-D – 15/11/2018 (Shift-II)

Ans : (a) LCM = HCF × 42
LCM + HCF = 602
HCF × 42 + HCF = 602
HCF (42 + 1) = 602
HCF = $\frac{602}{43} = 14$
So, First number × Second number = LCM × HCF
84 × Second number = 14 × 42 × 14
Second number = $\frac{14 \times 42 \times 14}{84} = 98$

127. Find the HCF and the LCM of 60 and 72 respectively.
(a) 16 and 360 (b) 12 and 260
(c) 12 and 360 (d) 10 and 360
RRB Group-D – 05/11/2018 (Shift-I)

Ans. (c) : On finding HCF by factorization method,
$60 = 2 \times 2 \times 3 \times 5$
$72 = 2 \times 2 \times 2 \times 3 \times 3$
HCF = 2×2×3 = 12
And finding the LCM by factorization method,
$60 = 2 \times 2 \times 3 \times 5$
$72 = 2 \times 2 \times 2 \times 3 \times 3$
LCM = 2 × 2 × 2 × 3 × 3 × 5 = 360
So, the HCF is 12 and the LCM is 360.

128. The HCF of two numbers is 110 and their product is 290400. The LCM of these numbers is:
(a) 1980 (b) 3960
(c) 1320 (d) 2640
RRB Group-D – 05/11/2018 (Shift-III)

Ans. (d) :
LCM × HCF = First number × Second number
LCM × 110 = 290400
LCM = $\frac{290400}{110}$
= 2640

129. The ratio of two numbers is 7:11 and their HCF is 8. What is the LCM of these numbers?
(a) 1540 (b) 924
(c) 616 (d) 308
RRB Group-D – 05/11/2018 (Shift-III)

Ans. (c) : According to the question,
Ratio of the numbers = 7:11
Let the numbers are 7x and 11x
HCF of 7x and 11x = x
HCF = 8
x = 8
So, the numbers = 56, 88.
∵ LCM × HCF = First number × Second number
LCM × 8 = 56 × 88
LCM = $\frac{56 \times 88}{8}$
= 56 × 11
= 616

130. The LCM and HCF of two numbers are 171360 and 14 respectively. If one from them is 2142 then find the other number.
(a) 12240 (b) 4128
(c) 1120 (d) 80
RRB Group-D – 25/10/2018 (Shift-II)

Ans : (c) LCM × HCF = First number × Second number
171360 × 14 = 2142 × Second number
Second number = $\frac{171360 \times 14}{2142}$
= 1120

131. If the LCM and HCF of two integers are 2184 and 2 respectively, and one from them is 78, then find the other number.
(a) 78 (b) 56
(c) 34 (d) 68
RRB Group-D – 24/10/2018 (Shift-III)

Ans. (b) : Let the other number is x.
Product of the numbers = LCM × HCF
78 × x = 2184 × 2
$x = \frac{2184 \times 2}{78} = 56$

132. If the product of two numbers is 3026 and their LCM is 89 then, their HCF is:

(a) 33 (b) 34
(c) 35 (d) 29

RRB NTPC 17.01.2017 Shift-1

Ans : (b) From formula,
Product of two numbers = LCM × HCF
$3026 = 89 \times HCF$
$HCF = \frac{3026}{89}$
$= 34$

133. If the product of two numbers is 4941 and their LCM is 81 then, what is their HCF?

(a) 60 (b) 59
(c) 35 (d) 61

RRB NTPC 17.01.2017 Shift-3

Ans : (d) From formula,
Product of two numbers = LCM × HCF
$4941 = 81 \times HCF$
$HCF = \frac{4941}{81} = 61$

134. The ratio of two number is 8:9 and their HCF is 6. Their LCM will be:

(a) 432 (b) 54
(c) 48 (d) 423

RRB NTPC 17.01.2017 Shift-3

Ans : (a) According to the question,
HCF of both numbers = 6
And their ratio is 8:9,
So, the First number = 8 × 6 = 48
Second number = 9 × 6 = 54
LCM × HCF = First number × Second number
$48 \times 54 = 6 \times LCM$
$LCM = \frac{48 \times 54}{6}$
$LCM = 432$

135. The LCM and HCF of two numbers are 616 and 2 respectively. If one is 22 then, find the other number.

(a) 87 (b) 56
(c) 116 (d) 36

RRB NTPC 03.04.2016 Shift : 3

Ans : (b) Let the other number is y.
From formula,
LCM × HCF = First number × Second number
$22 \times y = 616 \times 2$
$y = \frac{616 \times 2}{22}$
$y = 56$

136. The HCF and LCM of two numbers are 7 and 252 respectively. If one is 28 then, find the other number.

(a) 252 (b) 63
(c) 126 (d) 56

RRB NTPC 02.04.2016 Shift : 1

Ans : (b) ∵
LCM × HCF = First number × Second number
$7 \times 252 = 28 \times$ Second number
Second number $= \frac{7 \times 252}{28} = 63$

137. The HCF and LCM of two numbers are 3 and 2730 respectively. If one is 78 then, find the other number.

(a) 107 (b) 103
(c) 105 (d) 102

RRB NTPC 02.04.2016 Shift : 2

Ans : (c) According to the question,
HCF = 3
LCM = 2730
$N_1 = 78$
$N_2 = ?$
from formula,
$N_2 = \frac{LCM \times HCF}{N_1}$
$N_2 = \frac{2730 \times 3}{78}$
$N_2 = 105$

138. HCF of two numbers is 19 and their LCM is 665. If one is 95, then find the other number.

(a) 19 (b) 133
(c) 190 (d) 77

RRB NTPC 16.04.2016 Shift : 3

Ans : (b) Let the other number is N.
From formula,
LCM × HCF = First number × Second number
$95 \times N = 19 \times 665$
$N = \frac{19 \times 665}{95}$
$N = \frac{665}{5}$
$N = 133$

139. The LCM and HCF of two numbers are 3 and 2001 respectively. If one is 69, then find the other number.

(a) 58 (b) 87
(c) 29 (d) 23

RRB NTPC 10.04.2016 Shift : 3

Ans : (b) from question,
First number × Second number = LCM × HCF
69 × Second number = 2001×3
Second number $= \frac{6003}{69} = 87$
Hence, Second number = 87

140. The product of two numbers is 3276 and LCM is 63, then find their HCF.

(a) 54 (b) 51
(c) 53 (d) 52

RRB NTPC 07.04.2016 Shift : 2

Ans : (d) As we know,
First number × Second number = LCM × HCF
$3276 = 63 \times HCF$
$HCF = \frac{3276}{63} = 52$

141. Find the HCF and LCM of 16 and 24.
(a) 8, 48 (b) 4, 12
(c) 2, 24 (d) 4, 48

RRB NTPC 06.04.2016 Shift : 1

Ans : (a) On doing factorization,
$16 = 2\times2\times2\times2$
$24 = 2\times2\times2\times3$
So, HCF = 2 × 2 × 2 = 8
And LCM = 2 × 2 × 2 × 2 × 3 = 48

142. The ratio of two numbers is 21:29 and HCF is 8, then their LCM is:
(a) 4872 (b) 168
(c) 232 (d) 4782

RRB NTPC 26.04.2016 Shift : 2

Ans : (a) Let the numbers are 21x and 29x.
HCF = 8
So, first number = $21 \times 8 = 168$
Second number = $29 \times 8 = 232$
On, finding the LCM by factorization method,
$168 = 2\times2\times2\times3\times7$
$232 = 2\times2\times2\times29$
$LCM = 2\times2\times2\times3\times7\times29 = 4872$

143. The LCM of three numbers is 4752 and the HCF is 6. If two numbers are 48 and 66, find the third number.
(a) 54 (b) 56
(c) 58 (d) 52

RRB NTPC 29.04.2016 Shift : 3

Ans. (a) : Let the third number is x.
According to the question,
∵ HCF of all three numbers is 6.
The number in options which is divisible by 6 will be the third number.
So, from options, only 54 is divisible by 6.

144. If the product of two numbers is 3276 and their LCM is 63, then find their HCF.
(a) 32 (b) 76
(c) 52 (d) 53

RRB NTPC 30.04.2016 Shift : 3

Ans : (c) Product of numbers = LCM x HCF
$HCF = \frac{\text{Product of numbers}}{LCM}$
$HCF = \frac{3276}{63}$
HCF = 52

145. The ratio of two numbers is 5:7, and their HCF is 33, what is their LCM?
(a) 1155 (b) 165
(c) 231 (d) 1515

RRB NTPC 30.04.2016 Shift : 3

Ans : (a) According to the question,
HCF = 33
The ratio of numbers = 5:7
So, the first number = $5 \times 33 = 165$
And the second number = $7 \times 33 = 231$
$LCM = \frac{231 \times 165}{33} = 1155$

146. The HCF of two numbers is 12 and their LCM is 72. If one of them is 24, then the other number is:
(a) 48 (b) 60
(c) 36 (d) 72

RRB ALP & Tec. (21-08-18 Shift-II)

Ans : (c)
LCM × HCF = First number × Second number
$72 \times 12 = 24 \times$ Second number
Second number $= \frac{12 \times 72}{24}$
$= 36$

147. The ratio of two numbers is 14:9 and their HCF is 15. What is the LCM of these numbers?
(a) 1890 (b) 2520
(c) 1260 (d) 630

Ans. (a) : Let the numbers are 14x and 9x
HCF = 15
So, the first number = 14×15=210
The second number = $9 \times 15 = 135$
∵ LCM × HCF = First number × Second number
$LCM \times 15 = 210 \times 135$
$LCM = \frac{210 \times 135}{15}$
$= 1890$

148. The product of two numbers is 30. If one from them is 1.25, then what is the other number?
(a) 18 (b) 24
(c) 20 (d) 21

RRB Paramedical Exam – 21/07/2018 (Shift-II)

Ans : (b) Let the other number is x.
According to the question,
$1.25 \times x = 30$
$x = \frac{30}{1.25} = \frac{3000}{125}$
x = 24

Type - 7

149. Find the ratio of the LCM and HCF of the numbers 99 and 15.

(a) 165 : 1 (b) 3 : 55
(c) 165 : 3 (d) 1 : 165

RRB JE - 28/05/2019 (Shift-II)

Ans : (a) LCM of 99, 15

3	99,15
3	33,5
5	11,5
11	11,1
	1,1

LCM = 3 × 3 × 5 × 11 = 495
HCF of 99, 15
99 = 3×3×11
15 = 3×5
HCF = 3

So, the required ratio = $\frac{LCM}{HCF} = \frac{495}{3} = 165 : 1$

Type - 7

150. Two numbers are in a ratio 4:5. Their LCM is 180. Find their sum.

(a) 70 (b) 90
(c) 72 (d) 81

RRB JE - 24/05/2019 (Shift-III)

Ans : (d) Let the numbers are 4x and 5x.
The LCM of 4x and 5x = 20x
According to the question,
20 x = 180
x = 9
So, the numbers are 36 and 45.
Hence, the required sum = 36 + 45 = 81

151. Which number is a factor of all numbers?

(a) 2 (b) 1
(c) -1 (d) 0

RRB JE - 25/05/2019 (Shift-II)

Ans : (b) 1 is a factor of all numbers.
Example-
12 = 1 × 2 × 2 × 3
15 = 1 × 3 × 5

152. Two numbers are in a ratio 4:5 and their LCM is 100. Find their difference.

(a) 20 (b) 25
(c) 5 (d) 10

RRB RPF SI -11/01/2019 (Shift-III)

Ans : (c) Let the numbers are 4x and 5x.
According to the question,
The LCM of 4x and 5x = 20x
20x = 100
x = 5
So, the required difference = 5x – 4x = x = 5

153. If two numbers, whose HCF is 9, are in a ratio 5:7, then find their difference.

(a) 12 (b) 18
(c) 8 (d) 24

RRB JE - 28/06/2019 (Shift-III)

Ans. (b) Let the numbers are 5x and 7x.
HCF of 5x and 7x = x
∴ x = 9
So, the required difference = 7x – 5x = 2x
= 2 × 9 = 18

154. The ratio of two numbers is 5:6 and their LCM is 150. Find their sum.

(a) 50 (b) 55
(c) 60 (d) 54

RRB Group-D – 16/10/2018 (Shift-III)

Ans : (b) Let the numbers are 5x and 6x respectively.
LCM of 5x and 6x = 30x
According to the question,
30x = 150
x = 5
So, the numbers = 25 and 30
So, the required sum = 25 + 30 = 55

155. The LCM of two numbers is 120 and the sum of their squares is 289. The sum of these numbers is:

(a) 23 (b) 28
(c) 32 (d) 26

RRB Group-D – 22/10/2018 (Shift-III)

Ans : (a) Let the numbers are a and b.
$a^2 + b^2 = 289$
ab = 120
a = 15
b = 8
a + b = 15 + 8 = 23

156. The division of two numbers gives 6 and their product is 96. Find the product of the sum and the difference of these numbers.

(a) 540 (b) 560
(c) 592 (d) 9180

RRB Group-D – 24/10/2018 (Shift-II)

Ans. (b) Let the numbers are x and y.
According to the question,

$$\frac{x}{y} = 6 \Rightarrow x = 6y - - - - - (i)$$

$$xy = 96 - - - - - (ii)$$

On putting the value of equation (i) in equation(ii),

$$6y.y = 96$$

$$6y^2 = 96$$

$$y^2 = 16$$

$$y = 4$$

$$\therefore x = 6 \times 4 = 24$$

So, the required value,

$$(x+y)(x-y) = (24+4) \times (24-4)$$

$$= 28 \times 20 = 560$$

157. Which is the second greatest factor of 56 and 84?

(a) 23 (b) 18
(c) 14 (d) 24

RRB Group-D – 26/10/2018 (Shift-III)

Ans : (c) On doing factorization,
$56 = 2 \times 2 \times 2 \times 7$
$84 = 2 \times 2 \times 3 \times 7$
The first greatest factor = $7 \times 2 \times 2 = 28$
The second greatest factor = $7 \times 2 = 14$

158. What is the smallest number with 7 factors exactly?

(a) 100 (b) 36
(c) 64 (d) 16

RRB Group-D – 03/12/2018 (Shift-III)

Ans. (c) The number of factors of $a^x \times b^y \times c^z$ $\cdot (x+1)\times(y+1)\times(z+1)$ where a, b, c are prime numbers.
So,
100 = The number of factors of $2^2\times5^2 = (2+1)(2+1) = 9$
36 = The number of factors of $2^2\times3^2 = (2+1)(2+1) = 9$
64 = The number of factors of $2^6 = (6+1) = 7$
16 = The number of factors of $2^4 = (4+1) = 5$
So, it is clear that the required smallest number 7 factors exactly is 64.

159. The LCM of two numbers is 78. And the ratio of these numbers is 2:3. Find the sum of these numbers.

(a) 60 (b) 26
(c) 65 (d) 39

RRB NTPC 30.03.2016 Shift : 2

Ans : (c) Let the numbers are 2x and 3x.
So, the LCM = 78
$2 \times 3 \times x = 78$
$x = 13$
So, the required sum = $(5x) = 5\times13 = 65$

160. What is the highest common factor of 360 and 450?

(a) 90 (b) 45
(c) 10 (d) 9

RRB NTPC 29.03.2016 Shift : 3

Ans : (a) Note- The greatest factor of two numbers is their HCF.
So, on finding the HCF by division method,

```
360)450(1
    360
     90)360(4
        360
        ×××
```

So, the HCF is 90.

161. If P is the largest number which, when divides 60, 150 and 285, gives the same remainder in each case, then find the sum of digits of p.

(a) 7 (b) 5
(c) 4 (d) 9

RRB NTPC 19.04.2016 Shift : 3

Ans : (d) The required number = The HCF of (150 – 60), (285 – 150) and (285 – 60)
$90 = 2 \times \underline{3 \times 3 \times 5}$
$135 = 3 \times \underline{3 \times 3 \times 5}$
$225 = \underline{3 \times 3 \times 5} \times 5$
HCF = $3 \times 3 \times 5 = 45$
So, the required sum = $4 + 5 = 9$

162. The ratio of three numbers is 2:3:6 and their HCF is 45. Find the sum of the numbers.

(a) 405 (b) 455
(c) 495 (d) 525

RRB NTPC 18.04.2016 Shift : 2

Ans : (c) The given ratio = 2:3:6
And HCF = 45
$\therefore$ Numbers $\Rightarrow 2 \times 45 = 90$
$\Rightarrow 3 \times 45 = 135$
$\Rightarrow 6 \times 45 = 270$
So, the required sum = 90 + 135 + 270 = 495

163. The LCM of two numbers is 66. The ratio of the numbers is 2:3. The sum of the numbers is:

(a) 60 (b) 55
(c) 50 (d) 65

RRB NTPC 06.04.2016 Shift : 1

Ans : (b) The ratio of the numbers is 2:3.
Let the numbers are 2x and 3x.
The LCM of 2x and 3x = $2\times3\times x = 6x$
According to the question,
$6x = 66 \Rightarrow x = 11$
So, the required sum = $2x + 3x = 5x = 5\times11 = 55$

164. The LCM of two numbers is 48. The ratio of both the numbers is $1:\frac{2}{3}$. Find the sum of the numbers.

(a) 60 (b) 40
(c) 20 (d) 45

RRB NTPC 19.01.2017 Shift : 2

Ans : (b) Let the numbers are 3x and 2x.
According to the question,
LCM = 48
$6x = 48$
$x = 8$
So, the numbers = 24, 16
So, the required sum = 24 + 16 = 40

165. The HCF of two numbers is 16 and their difference is 16. Find the number.

(a) 64, 80 (b) 72, 88
(c) 80, 100 (d) 96, 120

RRB NTPC 29.04.2016 Shift : 2

Ans : (a) Solve this type of questions from options.
From option (a),
The difference of 80 and 64 = 16
And HCF of 64 and 80 is also 16.
From option (b),
The difference of 88 and 72 is 16 but their HCF is not 16.
From option (c),
The difference of 80 and 100 is 20.
From option (d),
The difference of 96 and 120 is 24.

166. The sum of two numbers is 30 and their LCM is 25. Find the larger number.

(a) 55 (b) 25
(c) 1 (d) 15

RRB NTPC 30.04.2016 Shift : 1

Ans.: (b) Let the larger number is x and greater number is y.
According to the question,

$x + y = 30$ ……(I)

And the LCM of x and y = 25,
From given options,
If x = 25, then y = 5 …… from option (b)
Then, the LCM of 25 and 5 = 25
And the sum = 25 + 5 = 30

Type - 9

167. What is the length of the side of the largest square tile, which is used for constructing the floor of tile of 13.92m length and 5.22m breadth.

(a) 58 cm (b) 1 m 16 cm
(c) 1 m 74 cm (d) 87 cm

RRB Group-D – 24/10/2018 (Shift-III)

Ans. (c) : The side of the largest square tile = HCF of 13.92m and 5.22m,
On finding the HCF by division method,

522)1392 (2
1044
348)522 (1
348
174)348 (2
348
×××

So, the HCF is 174.
Hence, the length of the side = 174cm = 1m 74cm,

Type - 10

168. Four alarm clocks ring at intervals of 3, 4, 5 and 8 minutes respectively. If they are rung together at 6:00 am, then what time will they ring together?

(a) 8:00 am (b) 10:00 am
(c) 12:03 pm (d) 09:00 am

RRB RPF SI -13/01/2019 (Shift-III)

Ans : (a)
LCM of 3, 4, 5 and 8 = $2 \times 2 \times 3 \times 5 \times 2 = 120$ min

$6 + \frac{120}{60} = 6 + 2 = 8$

So, at 8:00 am they will ring together.

169. Three bells ring at intervals of 10, 15 and 20 minutes. If they are rung together at 5:00 pm, what time will they ring together?

(a) 5:20 pm. (b) 5:30 pm.
(c) 5:40 pm. (d) 6:00 pm.

RRB RPF Constable -24/01/2019 (Shift-III)

Ans : (d) The LCM of 10, 15 and 20 = 60
The required time = 5:00 pm + 60 min
= 5:00 pm + 1 hour = 6:00 pm
So, the required time is 6:00 pm.

170. Alarms from three different clocks ring after 2, 4 and 6 hours respectively. If the clocks are started at the same time, how many times do the alarms ring simultaneously in 3 days?

(a) 6 (b) 3
(c) 2 (d) 9

RRB Group-D – 27/09/2018 (Shift-I)

Ans. (a) The time taken by all three clocks to ring together again
LCM of 2,4 and 6 = 12 hours
The number of alarms ring in three days (72 hours) = $\frac{72}{12} = 6$ times.

171. Traffic lights at any cross section change at 48 sec, 72 sec and 108 sec. If they change together at 9:15:00 am, when will they change together again next time?

(a) 9:22:12 a.m. (b) 9:21:07 a.m.
(c) 9:24:27 a.m. (d) 9:23:17 a.m.

RRB Group-D – 05/10/2018 (Shift-I)

Ans. (a) : LCM of 48 sec, 72 sec and 108 sec

2	48,	72,	108
2	24,	36,	54
2	12,	18,	27
2	6,	9,	27
3	3,	9,	27
3	1,	3,	9
3	1,	1,	3
	1,	1,	1

so, the LCM = 2×2×2×2×3×3×3 = 432
= 432 sec
= 7 min 12 sec
So, the traffic lights will change together next time =
9:15:00 + 0: 07:12
= 9:22:12 a.m.

172. Three bells ring at intervals of 15, 20 and 30 minutes respectively. If they ring together at 11:00 am, then when will they ring together again?

(a) 11.30 a.m. (b) 12 noon
(c) 12.30 p.m (d) 1.00 p.m

RRB NTPC 28.03.2016 Shift : 1

Ans : (b) LCM of 15, 20 and 30 = 60
So, all the three bells ring together at the interval of 60 minutes = 1 hour.
So, the bells will ring together at 11:00 + 1:00 = 12:00 noon.

173. In a temple, four bells ring together at intervals of 12, 16, 24 and 36 minutes respectively. If they start ringing at regular intervals from 6:00 an when will they ring together again?

(a) 8 : 24 am (b) 5 : 24 am
(c) 7 : 24 am (d) 6 : 24 am

RRB Group-D – 31/10/2018 (Shift-I)

Ans : (a) LCM of 12, 16, 24 and 36,

2	12,	16,	24,	36
2	6,	8,	12,	18
2	3,	4,	6,	9
2	3,	2,	3,	9
3	3,	1,	3,	9
3	1,	1,	1,	3
	1,	1,	1,	1

LCM = 2×2×2×2×3×3 = 144

So, 144 min = 2 hours and 24 minutes

So, the first interval = 6:00 + 2:24=8:24am

So, at 8:24 am the bells will ring together.

174. Four bells ring at intervals of 16, 24, 36 and 42 minutes respectively. If they were last ring together at 6:00 am, then after how many minutes will they ring together again?

(a) 842 minute (b) 964 minute
(c) 886 minute (d) 1008 minute

RRB NTPC 18.01.2017 Shift : 2

Ans : (d) The bells will ring together again = LCM of all numbers

LCM of 16, 24, 36 and 42.

2	16,	24,	36,	42
2	8,	12,	18,	21
2	4,	6,	9,	21
2	2,	3,	9,	21
3	1,	3,	9,	21
3	1,	1,	3,	7
7	1,	1,	1,	7
	1,	1,	1,	1

LCM = 2 × 2 × 2 × 2 × 3 × 3 × 7 = 1008

Hence, after 1008 minutes they will ring together.

175. Three clocks are designed to alarm in every hour, two hours and three hours respectively. If they all alarmed together three hours before, then after how many hours will they next alarm together?

(a) 3 hours (b) 6 hours
(c) 2 hours (d) 1 hours

RRB NTPC 27.04.2016 Shift : 3

Ans : (a) The clocks will ring together = LCM of 1, 2 and 3 = 6

So, they will ring together after 6 hours again.

Hence, after 3 hours later they will alarmed together

176. The traffic lights at four different road crossings change after every 15 sec, 18 sec, 27 sec and 30 sec respectively. If they all change simultaneously at 6:10:00 hours, then at what time will they again change simultaneously?

(a) 6:14:30 hours (b) 6:40:00 hours
(c) 6:14:00 hours (d) 10:40:00 hours

RRB NTPC 16.04.2016 Shift : 3

Ans. : (a)

Traffic lights will again change simultaneously

= LCM of 15, 18, 27 and 30

On finding the LCM by common division method,

2	15,	18,	27,	30
3	15,	9,	27,	15
3	5,	3,	9,	5
3	5,	1,	3,	5
5	5,	1,	1,	5
	1,	1,	1,	1

LCM = 2 × 3 × 3 × 3 × 5 = 270

So, after 270 seconds = 4 min 30 sec,

= 6:10:00 + 00:04:30 = 6:14:30

177. Three alarms ring at intervals of 10, 15 and 30 minutes respectively. If they ring together at 9:00 am, then when will they ring together again?

(a) 9:00 am (b) 10:00 am
(c) 10:30 am (d) 11:00 am

RRB NTPC 30.04.2016 Shift : 1

Ans : (a) LCM of 10, 15 and 30

2	10	15	30
3	5	15	15
5	5	5	5
	1	1	1

LCM = 2 × 3 × 5 = 30

So, all the three bells will ring together again after 30 minutes.

Hence, the required time is 9:00 + 30 = 9:30 am

178. Three clocks are designed to ring after every 1 hour, 2 hours and 3 hours respectively. If they ring together then after how many hours will they ring together?

(a) 3 hours (b) 6 hours
(c) 4 hours (d) 12 hours

RRB NTPC 29.04.2016 Shift : 2

Ans : (b) According to the question,

The interval for ringing three clocks' alarm together = LCM of 1 hour, 2 hours and 3 hours = 6

So, they will ring the alarm together after 6 hours.

179. Three bells ring at intervals of 15, 30 and 45 minutes respectively. At what time will they ring together again, if they rang simultaneously at 8.00 AM?

(a) 8.30 AM (b) 9.30 AM
(c) 9.00 AM (d) 8.45 AM

RRB ALP & Tec. (31-08-18 Shift-III)

Ans : (b) All the three bells will ring together again = LCM of 15, 30 and 45
On finding the LCM by common division method,

2	15, 30, 45
3	15, 15, 45
3	5, 5, 15
5	5, 5, 5
	1, 1, 1,

LCM = $2 \times 3 \times 3 \times 5 = 90$ min
So, the required time = 8:00 + 90 min = 9:30 AM

Type - 11

180. How many factors of 512 are perfect squares?
(a) 6 (b) 4
(c) 3 (d) 5
RRB Group-D – 28/09/2018 (Shift-I)

Ans : (d) On doing the factors of 512
= 1, 2, 4, 8, 16, 32, 64, 128, 256, 512
= 1, 4, 16, 64, 256 are perfect squares
So, the total number of perfect squares factors is 5.

181. Which is the smallest positive integer or natural number, when divides 1920 so that the number of factors of quotient is odd?
(a) 40 (b) 10
(c) 20 (d) 30
RRB Group-D – 12/12/2018 (Shift-I)

Ans. (d) From options,
$\frac{1920}{40} = 48 = \text{Number of factors in } 2^4 \times 3$
$= (4+1)(1+1) = 10(\text{Even})$
$\frac{1920}{10} = 192 = \text{Number of factors in } 2^6 \times 3$
$= (6+1)(1+1) = 14(\text{Even})$
$\frac{1920}{20} = 96 = \text{Number of factors in } 2^5 \times 3$
$= (5+1)(1+1) = 12(\text{Even})$
$\frac{1920}{30} = 64 = \text{Number of factors in } 2^6$
$= (6+1) = 7(\text{Odd})$

182. How many factors of the number $2^{10} \times 3^6 \times 5^3 \times 7^5$ are divisible by 2160?
(a) 180 (b) 336
(c) 504 (d) 560
RRB Group-D – 11/12/2018 (Shift-III)

Ans : (c) Factors of 2160 = $2^4 \times 3^3 \times 5^1$
Let the total factors are n.
$$n = \frac{2^{10} \times 3^6 \times 5^3 \times 7^5}{2^4 \times 3^3 \times 5^1}$$
$n = 2^6 \times 3^3 \times 5^2 \times 7^5$
So, the total number of factors = (6+1)(3+1)(2+1)(5+1)
= 7 × 4 × 3 × 6 = 504

Type - 12

183. Find the smallest number that is exactly divisible by each of the numbers 12, 15, 20 and 27.
(a) 540 (b) 510
(c) 530 (d) 520
RRB JE - 26/05/2019 (Shift-II)

Ans : (a) The required number = LCM of 12, 15, 20 and 27
On, finding the LCM by common division method,

3	12,15,20,27
5	4, 5, 20, 9
4	4, 1, 4, 9
9	1, 1, 1, 9
	1, 1, 1, 1

LCM = 3 × 5 × 4 × 9 = 540
So, the required number is 540.

184. Find the smallest three-digit number that is exactly divisible by 12, 15 and 24.
(a) 180 (b) 120
(c) 360 (d) 240
RRB JE - 26/05/2019 (Shift-III)

Ans : (b)
The required number = LCM of 12, 15 and 24
On finding the LCM by factorization method,
$12 = 2 \times 2 \times 3$
$15 = 3 \times 5$
$24 = 2 \times 2 \times 2 \times 3$
$\text{LCM} = 2 \times 2 \times 2 \times 3 \times 5 = 120$

185. Three cans of milk are of 36 liters, 45 liters and 72 liters measurement. Find the greatest measurement, which can measure all of them exactly.
(a) 15 liter (b) 8 liter
(c) 7 liter (d) 9 liter
RRB JE - 28/05/2019 (Shift-I)

Ans : (d) The required measurement = HCF of 36, 45 and 72
On finding the HCF by division method,

$36\overline{)45}\,(1$ $\quad$ $9\overline{)72}\,(8$
$\underline{36}$ $\quad$ $\underline{72}$
$9\overline{)36}\,(4$ $\quad$ $\times\times$
$\underline{36}$
$\times\times$

So, the HCF is 9.

186. Find the largest number that will divide exactly the product of four consecutive integers.
(a) 12 (b) 8
(c) 6 (d) 24
RRB JE - 02/06/2019 (Shift-II)

Ans. (d) The product of four consecutive integers

$= 2 \times 3 \times 4 \times 5 = 120$

Which is exactly divisible by 12, 6, 8 and 24. But 24 is the largest of all.

187. Find the smallest perfect cube number that is exactly divisible by 2, 3, 4 and 6.

(a) 1728 (b) 360
(c) 216 (d) 512

RRB JE - 01/06/2019 (Shift-I)

Ans : (c) LCM of 2, 3, 4 and 6 = The required number

On, finding the LCM by common division method,

2	2,3,4,6
2	1, 3,2,3
3	1, 3,1, 3
	1, 1,1, 1

LCM = $2 \times 2 \times 3 = 12$

So, the required perfect cube number $= 2\times2\times2\times3\times3\times3$
$= 216$

188. Find the greatest 3-digit number, which is exactly divisible by 12, 15 and 20.

(a) 115 (b) 240
(c) 180 (d) 120

RRB JE - 27/06/2019 (Shift-I)

Ans : (d) On finding the LCM of 12, 15 and 20

2	12	15	20
2	6	15	10
3	3	15	5
5	1	5	5
	1	1	1

LCM = $2 \times 2 \times 3 \times 5 = 60$

The smallest three digit number is 100.

So,

60) 100 (1
60
40

So, the required number = 100 + (60 – 40) = 120

189. What is the largest 4-digit number, divisible by 6, 9, 12, 13 and 15?

(a) 9326 (b) 9100
(c) 9360 (d) 9240

RRB Group-D – 11/12/2018 (Shift-II)

Ans : (c) On finding the LCM of 6, 9, 12, 13 and 15

2	6,	9,	12,	13,	15
2	3,	9,	6,	13,	15
3	3,	9,	3,	13,	15
3	1,	3,	1,	13,	5
5	1,	1,	1,	13,	5
13	1,	1,	1,	13,	1
	1,	1,	1,	1,	1

LCM = $2\times2\times3\times3\times5\times13 = 2340$

The largest four digit number = 9999

2340)9999(4
9360
639

$\therefore$

So, the required number = 9999 – 639 = 9360

190. Find the number between 300 and 500 which will be exactly divisible by 6, 8, 10 and 12:

(a) 320 (b) 340
(c) 490 (d) 360

RRB Group-D – 23/10/2018 (Shift-I)

Ans. (d) LCM of 6, 8, 10 and 12,

$6 = 2 \times 3$

$8 = 2 \times 2 \times 2$

$10 = 2 \times 5$

$12 = 2 \times 2 \times 3$

LCM $= 2 \times 2 \times 2 \times 3 \times 5 = 120$

So, the required number

= Multiples of 120 between 300 and 500.

= 360 and 480.

191. Find the least square number among the following, which is exactly divisible by 6, 9, 12, 13 and 15.

(a) 621000 (b) 456000
(c) 173000 (d) 152100

RRB Group-D – 08/10/2018 (Shift-I)

Ans. (d) From options, only 152100 is a perfect square, other numbers are not perfect squares.

LCM of 6, 9, 12, 13 and 15 = 2340

So,

$$\frac{152100}{2340} = 65 \text{ (Total divisible number)}$$

So, the required number is 152100 which is the square of 390.

192. Find the smallest square number which is exactly divisible by 4, 9 and 14.

(a) 1008 (b) 252
(c) 1764 (d) 504

RRB NTPC 17.01.2017 Shift-2

Ans : (c) LCM of 4, 9 and 14

2	4,	9,	14
2	2,	9,	7
7	1,	9,	7
3	1,	9,	1
3	1,	3,	1
	1,	1,	1

LCM = $2 \times 2 \times 3 \times 3 \times 7 = 252$

252 is not a square number. But its multiple 1764 is a square number.

So, from options 1764 is the required number.

193. What is the smallest 5-digit number which is exactly divisible by 12, 24, 48, 60 and 96?

(a) 10000 (b) 10024
(c) 10160 (d) 10080

RRB NTPC 03.04.2016 Shift : 1

Ans : (d) LCM of given numbers,

2	12,	24,	48,	60,	96
2	6,	12,	24,	30,	48
2	3,	6,	12,	15,	24
2	3,	3,	6,	15,	12
2	3,	3,	3,	15,	6
3	3,	3,	3,	15,	3
5	1,	1,	1,	5,	1
	1,	1,	1,	1,	1

LCM = 2×2×2×2×2×3×5 = 480

The smallest 5-digit number = 10000

480)10000(20
9600
400

So, the required number = 10000 + (480 – 400)
= 10080

194. Which is the largest 3-digit number which is exactly divisible by 15, 25 and 30?

(a) 900 (b) 930
(c) 960 (d) 975

RRB NTPC 28.03.2016 Shift : 2

Ans :(a) LCM of 15, 25 and 30 = 150

The largest three digit number = 999

150) 999 (6
900
×99

So, the required number = 999 – 99 = 900

195. Find the largest four digit number which is totally divisible by 12, 20, 32 and 44.

(a) 5200 (b) 4719
(c) 6800 (d) 5280

RRB NTPC 12.04.2016 Shift : 2

Ans : (d) LCM of 12, 20, 32 and 44

2	12	20	32	44
2	6	10	16	22
	3	5	8	11

$LCM = 2 \times 2 \times 3 \times 5 \times 8 \times 11 = 5280$

The largest 4-digit number = 9999

5280)9999(1
5280
4719

So, the required number = 9999 – 4719 = 5280

196. Find the largest number which is totally divisible by 15, 18, 27 and 30.

(a) 870 (b) 900
(c) 810 (d) 780

RRB NTPC 12.04.2016 Shift : 3

Ans : (c) LCM of 15, 18, 27 and 30

$15 = 3 \times 5$

$18 = 2 \times 3 \times 3$

$27 = 3 \times 3 \times 3$

$30 = 2 \times 3 \times 5$

$LCM = 2 \times 3 \times 3 \times 3 \times 5 = 270$

So, the required number should be divisible by 270.

From options, only 810 is divisible by 270.

197. Which is the smallest number, when doubled is exactly divisible by 4, 6, 9, 12 and 14?

(a) 126 (b) 252
(c) 504 (d) 63

RRB NTPC 19.01.2017 Shift : 2

Ans : (a) Let the required number is x.

According to the question,

LCM of 4, 6, 9, 12 and 14

$4 = 2 \times 2$

$6 = 2 \times 3$

$9 = 3 \times 3$

$12 = 2 \times 2 \times 3$

$14 = 2 \times 7$

$LCM = 2 \times 2 \times 3 \times 3 \times 7 = 252$

So, 2x = 252

x = 126

198. Find the smallest number which, when doubled is exactly divided by 14, 35, 28 and 91.

(a) 14 (b) 1820
(c) 910 (d) 1260

RRB NTPC 22.04.2016 Shift : 3

Ans : (c) According to the question,

LCM of 14, 35, 28 and 91

2	14,	35,	28,	91
7	7,	35,	14,	91
	1,	5,	2,	13

LCM = 2 × 7 × 5 × 2 × 13 = 1820

So, the required number = $\frac{1820}{2} = 910$

199. Find the smallest number which, when divided by 10, 15, 20 and 25, gives remainder 3 in each case.

(a) 300 (b) 303

(c) 306 (d) 309

RRB NTPC 26.04.2016 Shift : 1

Ans : (b) LCM of 10, 15, 20 and 25,

2	10	15	20	25
2	5	15	10	25
3	5	15	5	25
5	5	5	5	25
5	1	1	1	5
	1	1	1	1

LCM = 2 ×2 × 3 × 5 × 5 = 300

So, the required number = 300 + remainder

= 300 + 3 = 303

200. Find the smallest number which is exactly divisible by 6, 8, 12 and 16.

(a) 48 (b) 24

(c) 64 (d) 80

RRB NTPC 27.04.2016 Shift : 3

Ans : (a) The required number = LCM of 6, 8, 12 and 16

On finding the LCM by common division method,

2	6,	8,	12,	16
2	3,	4,	6,	8
2	3,	2,	3,	4
2	3,	1,	3,	2
3	3,	1,	3,	1
	1,	1,	1,	1

LCM = 2 × 2 × 2 × 2 × 3 = 48

201. Find the largest number which, when divides 1250 and 1615, gives remainder 4 and 5 respectively.

(a) 13 (b) 14

(c) 16 (d) 18

RRB NTPC 28.04.2016 Shift : 1

Ans : (b) According to the question,

1250 – 4 = 1246

1615 – 5 = 1610

HCF of 1246 and 1610,

```
1246)1610(1
     1246
      364)1246(3
          1092
           154)364(2
               308
                56)154(2
                   112
                    42)56(1
                       42
                       14)42(3
                          42
                          ××
```

HCF = 14

202. What is the least consisting of five digits that is exactly divisible by 12, 18, 20 and 25?

(a) 10000 (b) 10800

(c) 11250 (d) 10680

RRB ALP & Tec. (31-08-18 Shift-II)

Ans. (b) : On finding the LCM of given numbers,

2	12	18	20	25
2	6	9	10	25
3	3	9	5	25
3	1	3	5	25
5	1	1	5	25
5	1	1	1	5
	1	1	1	1

LCM = 2×2×3×3×5×5 = 900

900 is the required three-digit,

so the product of 900 which is smallest

5-digit number = 900 × 12 = 10800

203. Find the smallest square number from among the given option which is divisible by each of 8, 15 and 20.

(a) 3600 (b) 6400

(c) 14400 (d) 4900

RRB ALP & Tec. (13-08-18 Shift-I)

Ans : (a) LCM of 8, 15 and 20.

8 = 2 × 2 × 2

15 = 3 × 5

20 = 2 × 2 × 5

LCM = 2 × 2 × 2 × 3 × 5 = 120

From options, only (a) 3600 is such which is square of 60 and divisible by 120 which is LCM of 8, 15 and 20.